

Network Form Instructions

SW12-605

**Implementation
Startup & Reconfiguration - 1**

Network Form Instructions

**SW12-605
Release 684**

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About This Publication

The information in this publication will assist you to understand and document the following system-wide and network configuration aspects for a **TotalPlant** Solution (TPS) system. TPS is the evolution of TDC 3000^X.

The naming of units, areas, and consoles

Assignment of nodes to the network

The configuration of volumes for the History Module

This publication does not explain how to enter configuration information into the TPS System. That information is provided in *Network Data Entry* manual located in the *Implementation/Startup & Reconfiguration - 1* binder.

Before using this publication you should be familiar with the contents of these publications:

Title	Binder
Configuration Data Collection Guide	Implementation/Startup & Reconfiguration - 2
Engineer's Reference Manual	Implementation/Startup & Reconfiguration - 2

This publication supports TPS software release 684.

NOTE

With TPN R684, Enhanced Universal Control Network (EUCN) is introduced. In EUCN, the traditional coaxial cable system is no longer utilized and is replaced with a Fault Tolerant Ethernet (FTE) based communication network.

Throughout the document, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms **EHPM**, **ENIM**, and **EUCN**, respectively.

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1 INTRODUCTION

1.1 PURPOSE

This publication provides instructions on how to fill in the *Network Forms* Local Control Network configuration forms. These forms are designed to make it easy to collect and organize data for configuration.

1.2 SCOPE

This publication explains how to fill out the forms in the publication *Network Forms*. This includes forms for the following:

- Unit Names (IDs/descriptors, up to 100 for each LCN)

- Area Names (Area names/descriptors, up to 10 for each LCN)

- Console Names (Console numbers/descriptors, up to 10 for each LCN)

- LCN Nodes

 - Number Assignment

 - Node-type Assignment (CM, HG, HM, AM, US, NIM, or NG) and details for each type

- System-Wide Values

 - System Descriptor (printer-page width and length, work-shift details, horn-annunciation policy, and access assignment to physical node commands)

- Volume Configuration (by pairs of History Module Nodes or "Node Pairs")

 - Node-type selection and dump data

 - Area Names/volume size (up to 10/Node Pair)

 - Checkpointing assignments

 - Continuous History

 - Unit-Group assignments (up to 60 Units and 150 Groups/Node Pair)

 - Archive Options (Save, Prearchive Hours)

- Process Journals

 - Unit Assignment (up to 36 Units/Node Pair)

 - Journal Size (for each journal)

 - Number of events to be journaled

- CL Storage Configuration (Hiway, number of files, and volume size)

- User Files Storage Volume (Virtual Volume ID, number of files, average file size; up to 15 volumes for each Winchester-disk drive)

1.3 PREREQUISITES

Do the following before filling in the configuration forms:

- Understand the fundamentals of **TotalPlant** Solution (TPS) system network configuration (refer to the *Configuration Data Collection Guide*).
- Collect the data for the TPS system that is to be configured.

1.4 REFERENCES

These publications are useful in Network configuration:

<u>Title</u>	<u>Binder</u>
Application ModuleX System Administration Manual	Application ModuleX
Configuration Data Collection Guide	Implementation/Startup & Reconfiguration - 2
Engineer's Reference Manual	Implementation/Startup & Reconfiguration - 2
Network Data Entry	Implementation/Startup & Reconfiguration - 1
Network Forms	Implementation/Configuration Forms

Make as many copies of the forms in *Network Forms* document as are needed. A sample of each form is included in this publication, next to instructions for filling it in.

2 GENERAL INSTRUCTIONS

2.1 OVERVIEW

Instructions in this publication describe the entries to be made on the Network configuration forms. These forms match the "screen forms" that are part of the **TotalPlant** Solution (TPS) system network configurator function. The instructions in this publication correspond to configuration data entries made through the Universal Station.

2.2 PARAMETER DISCLOSURE NOTATION

Instructions for filling in the forms are in a tabular format. Each line on the form has a number that corresponds to a number in the table for that form, where the choices and instructions are given. Some of the entries require additional entries. The relationship between the line(s) that require additional entries and the line(s) on which the additional entries are made is indicated by alphabetic suffixes. For example, Lines 8A, 8B, and 8C depend on the entries made in line 8.

2.3 DEFAULTS

Default values are values that are assigned to the data point by the system. When the user chooses not to make an entry in a particular line, the default value is used. The default value for each parameter is listed in these form instructions, and on the data-entry displays.

2.4 PICK AND PORT NOTATION

The forms have two types of entries:

1. Picks (also called targets)—For selecting options. All options are on the forms. For each such entry, circle one of the options.
2. Ports—For entering data. This form instruction gives the maximum number of characters and the set of valid characters; write in an alphanumeric character-string.

2.5 CONSIDERATIONS AND GUIDELINES

The considerations and guidelines in this publication are taken from the Engineer's Reference Manual. You should be familiar with that publication before filling in the Network Configuration Forms.

Network Configuration provides a foundation for all other TPS system configuration. If it is not properly planned and completed, it is possible that some configuration tasks would have to be done again. This could involve significant effort and inconvenience. With proper planning, reconfiguration can be minimized or eliminated.

2.5.1 Network Configuration File (NCF.CF)

For Network Configuration, data that is entered for LCN-based functions (Unit, Console and Area Names, LCN Nodes, System-Wide Values, and Volumes) is ultimately installed in the Network Configuration File (NCF.CF). Following are some considerations and guidelines that could minimize configuration errors.

2.5.1.1 Unit Names, Consoles Names, and Area Name Configuration

Unit Names, Console Names, and Area Names are internally used by the system to organize and access data in various operations. In configuration, the position of a name in a list (relative to other names in the list) is retained. If a list is rearranged, or if a name is added to or deleted from the list other than the end of the list, the external names used by the system are changed and the file must be recompiled, reloaded and installed.

Plan Ahead– Ensure that the lists of names are correct as configured and add to only the end of a list if it needs to be changed.

2.5.1.2 LCN Nodes Configuration

Entries on the Network configuration forms must match the hardware configuration of each LCN Node. Configuration of Universal Station Nodes also effectively defines consoles, which are identified by numbers that are unique to a console. Note that a Programmable Logic Controller Gateway (PLCG) is configured on a HG node.

If you purchased custom software (external load modules) or plan to use the background AM/CL software, you will need to refer to that documentation for essential values used on the second or third page of each LCN node configuration screen.

Plan Ahead– If you anticipate that other nodes and devices on consoles will be added to your system, include them in the current configuration to avoid having to shut down/start up parts of the system because of a change to the Network Configuration file (NCF.CF).

Before configuring HG, NIM, or NG Nodes, ensure that networks are sufficiently defined to permit assignment of identifying numbers.

2.5.1.3 System-Wide Values Configuration

Because changes to the Network Configuration file cause at least partial shutdown/restart, it is important to precisely define these values. Do not change revisions of printed-wire boards (last few pages of the System-Wide Values display) unless requested by Honeywell to do so.

2.5.1.4 Volume Configuration

Volume configuration defines how the Winchester disk(s) associated with each History Module (HM) is to be partitioned to store system and user volumes. The definition of disk partitioning is stored on the disk in the Local Network Volume Configuration File (for example, L01 NVCF.MM), which is created during initialization of HM number 1. When an HM is initialized, it erases any previous Local NVCF file and all the data in all volumes and directories that are stored on the disk. Any information on the disk(s) that you want to save should be copied to removable media (cartridge or floppy disks) before the HM is initialized with the new Network Configuration File (NCF.CF).

Plan Ahead— Use the following information to plan R500 HM partitioning:

General: Total space available on an HM with

- each type 3 Winchester drive = 136,900 kb
- SCSI compatible drives as follows:

	<u>Device</u>	<u>Type</u>	<u>Nominal Size</u>	<u>Formatted Space</u>
•	each ST 3283 drive =	4	210 MB	215,040 kb
•	each ST 3600 drive =	5	445 MB	454,130 kb
•	each ST 31230N drive =	6	875 MB	896,046 kb

A. User Volume Recommendations

1. For ease of identification, you may want to create separate user volumes for—
 - Intermediate Data Files (IDFs)
 - Control Language (CL) Source files
 - Exception Build (EB) Source files
 - Custom Graphic files
 - CL Custom-GDF (CDSG) files
 - Custom Software files

Alternatively, separate directories within a user volume can be created for any of these functions. The first user volume can be configured to have either file descriptors or fast access. We recommend that you configure it for fast access and use it to store custom files where call up time may be time critical.

2. Sizing recommendations/suggestions—Refer to the *Engineer's Reference Manual* (see References).

3 SPECIFIC INSTRUCTIONS

3.1 GENERAL

This section contains detailed instructions on how to fill out each of the Network configuration forms. The instructions are in tables that correspond on a line-by-line basis to the configuration forms.

Default choices on the forms are indicated by bold print. Default values are shown near the entry port.

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.2 UNIT NAMES CONFIGURATION FORM SW88-506

Table 3-1 contains instructions for Configuration Form SW88-507.

Table 3-1 — Entries for Unit Names Description Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
NOTE Unit Names as internally used are indexes to the characters as configured for Unit Names. If the list of Unit Names is changed so as to change its sequence, this change WILL NOT BE REFLECTED IN INTERNAL UNIT IDs such as Continuous History.			
1	UNIT ID	Enter two characters for each of up to 100 Units. Unit IDs cannot be changed or deleted in On-Line mode.	Unit IDs are user-defined and can include alpha characters and numeric characters; lower-case alpha characters are automatically converted to upper-case. A Unit ID cannot include an underbar (_), and cannot contain a space. Numeric Unit IDs can have a leading zero (for example, 03). Each Unit ID must be unique in the system; numeric Unit IDs "3" and "03" are not allowed in the same system. The characters SY are re-served and are not allowed as Unit ID.
	UNIT DESCRIPTION	Enter up to 24 characters for each Unit ID configured.	Unit description can include any character on the Engineering Keyboard; lower-case letters are automatically converted to upper case.

AREA NAMES CONFIGURATION FORM

EDIT AREA NAMES

PAGE 1

1 AREA NAME AREA DESCRIPTION

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References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.3 AREA NAMES CONFIGURATION FORM SW88-507

Table 3-2 contains instructions for Configuration Form SW88-507.

Table 3-2 — Entries for Area Names Description Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
1	AREA NAME	Enter up to eight characters for each of up to 10 areas.	Area names can include alphanumeric characters or underbar. Area names must be legal identifiers.
	AREA DESCRIPTION	– up to 24 characters for AREA DESCRIPTION	Area descriptions are defined by the user, and can include any characters on the Engineer's Keyboard. Lower-case letters are converted to upper case.

CAUTION

1. Enter Area Names consecutively (that is, no blank lines between names).
2. If you use short numbered area names like 1, 2, 3, and so on, we suggest that you put them on lines 1,2,3 so on, respectively, to avoid confusion. During an area change, the line number and area name for all configured areas appear side by side (for example, **01 1**). If you use the name Area01 on line one it appears as **01 Area01**; if you used the name A, it appears as **01 A**.

NOTE

Following a power dropout, certain recovery techniques are sometimes easier and faster if at least two Areas are configured. Therefore, Honeywell recommends that you enter at least two area names. If all 10 areas are not needed, we recommend that you enter one spare area name (you will not need to build a database for the spare area).

CONSOLE NAMES CONFIGURATION FORM

EDIT CONSOLE NAMES

PAGE 1

1	CONSOLE NUMBER	CONSOLE DESCRIPTION
	1	_____
	2	_____
	3	_____
	4	_____
	5	_____
	6	_____
	7	_____
	8	_____
	9	_____
	10	_____

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.4 CONSOLE NAMES CONFIGURATION FORM SW88-508

Table 3-3 contains instructions for Configuration Form SW88-508.

Table 3-3 — Instructions for Console Names Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
1	CONSOLE DESCRIPTION	Enter up to 24 characters for each of up to 10 consoles.	Console descriptions can consist of any characters on the Engineer's Keyboard. Lower-case letters are converted to upper case.

LCN NODES CONFIGURATION (REFERENCE) FORM

LCN NODE CONFIGURATION - SELECT DESIRED NODE

1

NODE NO.	NODE TYPE	REDUNDANT		NODE NO.	NODE TYPE	REDUNDANT		NODE NO.	NODE TYPE	REDUNDANT	
		N/P/S	NO.			N/P/S	NO.			N/P/S	NO.

PAGE 1

1	.	.	.	.	.	.	.	9	.	.	.	.	.	.	.	17	.	.	.	.	.	.	.	.	.	.
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27	.	.	.	.	.	.	.	35	.	.	.	.	.	.	.	43	.	.	.	.	.	.	.	.	.	.
28	.	.	.	.	.	.	.	36	.	.	.	.	.	.	.	44	.	.	.	.	.	.	.	.	.	.
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31	.	.	.	.	.	.	.	39	.	.	.	.	.	.	.	47	.	.	.	.	.	.	.	.	.	.
32	.	.	.	.	.	.	.	40	.	.	.	.	.	.	.	48	.	.	.	.	.	.	.	.	.	.

PAGE 3

49	.	.	.	.	.	.	.	57	.	.	.	.	.	.	.
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54	.	.	.	.	.	.	.	62	.	.	.	.	.	.	.
55	.	.	.	.	.	.	.	63	.	.	.	.	.	.	.
56	.	.	.	.	.	.	.	64	.	.	.	.	.	.	.

NOTES: Type = US or ^XS (Universal Station)
= HG (Hiwav Gatewav or PLCG)
= HM (History Module)
= AM or ^XM (Application
= CM (Computing Module)
= NIM (Network Interface
= NG (Network Gateway Module)

N/P/S = None, or
= Primarv. or
= Secondary

3.5 LCN NODES CONFIGURATION (REFERENCE) FORM SW88-609

Table 3-4 contains instructions for Configuration Form SW88-609, Page 1.

Table 3-4 — Entries for LCN Nodes Configuration (Reference) Form

LINE	NAME	FORM ENTRY	REMARKS
1	NODE NO.	No entry	Node No. is preprinted on the form. It is the physical node number (or node address) that is configured in hardware.
	NODE TYPE (See Note 2)	Enter as required: US II HG II HM II AM II CM II NIM II NG II US IV HG IV HM IV AM IV CM IV NIM IV NG IV	US = Universal Station or Universal Station ^X HG = Hiway Gateway or PLCG HM = History Module AM = Application Module or Application Module ^X CM = Computing Module NIM = Network Interface Module NG = Network Gateway
	REDUNDANT NODE – N/P/S	Enter N or P or S, and	N = no redundant node P = Primary node S = Secondary node
	– NO.	Enter 2-digit number.	NO. = number of redundant node (primary or secondary).
NOTES - Page 1 of this form is only for reference. No configuration data entry can be made through this display. Data that appears in the LCN Nodes Configuration display is actually entered through displays that correspond to the remaining pages in form SW88-609. - If a module uses an HMPU, HPK2, or K2LCN printed wire board, circle the type II version. If a module contains a K4LCN, circle the type IV version. - A Programmable Logic Controller Gateway (PLCG) is configured as a HG.			

LCN NODES CONFIGURATION FORM
(Continued)

UNIVERSAL STATION NODE DISPLAY

1 NODE

2 CONSOLE # STATION # DEFAULT AREA

3 Console Peripheral Device Numbers:

CARTRIDGE DISC NUMBERS |

FLOPPY DISC NUMBERS |

PRINTER NUMBER

TREND PEN NUMBER

4 STATION DEFAULT ACCESS LEVEL VIEW **OPER**

5 STATION DEFAULT LOAD PERSONALITY UP **OPER**

6 STATION OPTIONS

OPERATOR'S KEYBOARD TOUCH SCREEN ENGINEER'S KBD ANNUN DURING ENG FUNC

YES **NO**

YES **NO**

YES **NO**

YES **NO**

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.6 UNIVERSAL STATION NODE CONFIGURATION FORM SW88-609

Table 3-5 contains instructions for Configuration Form SW88-609, Page 2.

Table 3-5 — Instructions for Universal Station/Universal Station^x Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
UNIVERSAL STATION NODE DISPLAY			
1	NODE	Enter the node number to be configured.	Physical node number; refer to Table 3-4.
2	CONSOLE #	Enter a number from 1 to 10.	Assigns this Universal Station to a console.
	STATION #	Enter a number from 1 to 10.	Assigns unique number to this station.
	DEFAULT AREA	Enter one of the area names defined on Form SW88-507.	Defines default area data base to be loaded into the specified Universal Station.
3	Console Peripheral Device Numbers		
	– CARTRIDGE DISC NUMBERS	Enter one or two numbers from 1 to 20, unique within a console.	Assigns a logical-device number to each cartridge disk drive attached to this Universal Station.
	– FLOPPY DISC NUMBERS	Enter one or two numbers from 1 to 20, unique within a console. You cannot mix floppy and cartridge disks.	Assigns a logical-device number to each floppy disk drive attached to this Universal Station.
	– PRINTER NUMBER	Enter a number from 1 to 10, unique within a console (see NOTE at end of table).	Assigns a logical-device number to the printer attached to this US.
	– TREND PEN NUMBER	Enter a number from 1 to 10, unique within a console.	Assigns a logical-device number to the Trend Pen assigned to this console.
4	STATION DEFAULT ACCESS LEVEL (The level of access without a key)	Circle one: VIEW OPER	Select: –OPER for normal operator access to the operating displays called up at this Universal Station. –VIEW to configure this US for view-only access to operating displays.
5	STATION DEFAULT LOAD PERSONALITY	Circle one: UP OPER	Select the default personality to be loaded by Autoload Net, and the reload options.
6	STATION OPTIONS		
	OPERATORS KEYBOARD	Circle one: YES NO	Select YES if the station has an Operator Keyboard.
	TOUCH SCREEN	Circle one: YES NO	Select YES if the station has a touch screen. Also select YES for a UxS Station with a Mouse or Trackball.
	ENGINEER KEYBOARD	Circle one: YES NO	Select YES if the station has an Engineer's Keyboard

(Continued)

Table 3-5 — Instructions for Universal Station/Universal Station^X Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
UNIVERSAL STATION NODE DISPLAY			
6	ANNUN DURING ENG FUNC	Circle one: YES NO	Select YES to activate the annunciator when an alarm condition is detected while the Engineering Main Menu is being displayed or an EngineeringMain Menu function is being executed. Select NO if the annunciator is not to be activated when an alarm condition is detected while the Engineering Main Menu is being displayed or anEngineering Main Menu function is being executed.

LCN NODES CONFIGURATION FORM (Continued)

UNIVERSAL STATION NODE DISPLAY, Continued

EXTERNAL LOAD MODULE NAMES & ASSOCIATED PERSONALITY TYPES

6

NAME

PERS

A handwriting practice sheet with two columns of three rows each. Each row contains a solid top line, a dashed middle line, and a solid bottom line, with vertical dashed lines dividing the space into six columns.

OPR

UP

8 ADDITIONAL MODULE
MEMORY (WORDS)

9 MAXIMUM EXTERNAL
MODULE MEMORY (WORDS)

10 FURTHER EXTERNAL
DIRECTIVES?

YES NO

YES NO

```

11  EXTERNAL CUSTOM
    SCHEMATIC MEMORY
    (KILOWORDS)

```

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-5 contains instructions for Configuration Form SW88-609, Page 3. If you did not purchase custom software (External Load Modules) skip to the next section.

Table 3-5 — Instructions for Universal Station Configuration Form (Continued)

LINE	NAME	FORM ENTRY	REMARKS
UNIVERSAL STATION NODE DISPLAY, Continued			
6	NAME	Enter the name(s) of External Load Modules that use this node.	Get exact Load Module name from instructions with the custom software.
8	ADDITIONAL MODULE MEMORY (WORDS) OPR/UP	Enter size of any additional memory to be reserved for this node. Enter for each applicable personality.	Get this information from documentation that comes with the Load Module.
9	MAXIMUM EXTERNAL MODULE MEMORY (WORDS) OPR/UP	Enter maximum memory words to be allowed for External Load Modules for each personality.	If this information is not in documentation that comes with the Load Module, leave blank. It can be determined at data entry time.
10	FURTHER EXTERNAL DIRECTIVES? YES, NO- OPR/UP	Circle NO unless otherwise instructed by documentation with the custom software.	YES means an external Load directive file must be accessed when the node is loaded with this personality.
11	EXTERNAL CUSTOM SCHEMATIC MEMORY (KILOWORDS)	Enter a number or "ALL". ALL reserves all available memory above node minimum for schematics. If you have external load modules, do not enter ALL.	If you need to add more hardware memory to the US refer to the service manuals. Available memory must hold a US personality, optional load modules, and the largest schematic.

NOTE

As long as one printer is present within the console, you can make screen prints from US stations in that console that do not actually have a printer attached. To enable this function, configure a printer on US stations from which screen printing is desired. Enter a different printer number for each station.

After network configuration tasks are complete and the new NCF file is installed, you can use the Printer Assignment function on the Console Status display to direct screen prints to a printer that is present. The Peripheral Status for stations configured this way is "PRINTER NOT PRES".

With R681, NCF print option enables you to print NCF to a text file. Initiate a print using F4 function key from any NCF form to print the complete NCF as a text file. It will be stored in NET>&ASY by default with the file name as NCF.XX. If the default filename is not desired, specify a text file name using the extensions ".X?" or ".Y?" or ".Z?".

Where,
? can be any alphabet.

If &ASY has memory shortage, you can print the NCF.XX to a different user volume.

You can use the existing functionality to abort printing – simultaneously press the keys CTL + Break. The printing is aborted and the partially printed text file is retained with a comment PRINT OPERATION ABORTED.

For an NCF parameter with multiple values, the values are printed continuously separated by a space. If the number of characters exceeds 80, the next line begins with &\, indicating it is a continuation of the previous line.

Note that in large networks having more than 64 nodes, the NCF text file takes approximately 30 minutes to print.

LCN NODES CONFIGURATION FORM (Continued)

APPLICATION MODULE NODE

```

1 NODE                | | |
2 REDUNDANT MEMBER NODE ID | | |
3 STARTUP MODE        COLD    WARM    HOT    NOPROC

```

4 ASSIGNED UNIT LIST

[illegible]

USER MEMORY ALLOCATION

5	FUNCTIONAL ADJUSTMENTS:	Default
5A	# BACKGROUND CL TASKS	0
5B	# CONCURRENT DATA ACCESSES FROM BACKGROUND CL	0
5C	BACKGROUND TASK STACK SIZE	0
5D	CVB SIZE FOR FAST & SLOW POINT PROCESSORS	2000
5E	REDUNDANCY BUFFER INCREASE (# OF 32KW BLOCKS)	0
5F	INCLUDE INTERNETWORK POINT PROCESSOR	YES NO
5G	CVB SIZE FOR IPP	0
5H	USER MEMORY RESERVED (# OF 32KW BLOCKS)	0

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.7 APPLICATION MODULE NODE CONFIGURATION FORM SW88-609

Table 3-6 contains instructions for Configuration Form SW88-609, Page 4.

Table 3-6 — Instructions for Application Module Node Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
APPLICATION MODULE NODE DISPLAY (Use for AM or A^XM)			
1	NODE	Enter a number from 1 to 64.	Physical node number
2	REDUNDANT MEMBER NODE ID	Enter a number from 1 to 64.	Physical node number of the optional redundant AM. This number must be larger than the primary node number.
3	STARTUP MODE	Circle one: COLD WARM HOT NOPROC	Selects the default startup mode. Refer to the <i>Control Functions Reference Manual</i>
4	ASSIGNED UNIT LIST	Enter 1 to 62 Unit IDs in startup order (from Form SW88-506). Separate IDs with commas or spaces. For an AM with a redundant node the limit is 61.	Defines the Units that include the points assigned to this AM. A given Unit can be assigned to only one AM and/or one CM.
5 FUNCTIONAL ADJUSTMENTS - Most of the items below are for use with A ^X M or Background CL, CL/AM extensions and other specialized functions. If you have not written or purchased packages that use these specialized functions, skip them. These items can be entered later using the on-line reconfiguration process. Items 5F and 5G are for use with an optional Network Gateway. For A ^X M, see the <i>Application ModuleX System Administration</i> manual.			
5A	# BACKGROUND CL TASKS	Enter the number of CL back-ground tasks you want to use.	Range is 0 - 10. Memory used includes stack size. The larger the setting, the more AM memory required.
5B	# CONCURRENT DATA ACCESSES FROM B/GND CL	Enter the number of concurrent data access tasks allowed from back-ground CL.	Range is 0 - 4. It must equal at least 1 if the number of Background CL Tasks is > 0.
5C	BACKGROUND TASK STACK SIZE	Enter CL Background Task stack size (in words).	Range is 700 - 32000. Default is 15000.
5D	CVB SIZE FOR FAST & SLOW PT. PROCESSORS	Enter Current Value Buffer size for fast or slow point processor (in words).	Range is 2000 - 15000. Default is 2000. See Eng. Ref. Manual, AM Memory Size.
5E	REDUNDANCY BUFFER INCREASE (# OF 32 KW BLOCKS)	Enter number of additional 32 KW blocks in AM Redundancy Buffer (in words).	Range is 0 - 12; Default is 0. Redundancy Buffer increase plus user memory reserved must be <=12.
5F	INCLUDE INTER-N/W POINT PROC.	Select: YES or NO	Yes permits internetwork access from assigned ports.
5G	CVB SIZE FOR IPP	Enter the Current Value Buffer size (in words) for IPP.	Range is 2000 - 15000. IPP = Internetwork Point Processing.
5H	USER MEMORY RESERVED (# OF 32 KW BLOCKS)	Enter # 32 KW blocks reserved for future use.	Range is 0 - 12; Default is 0. Redundancy buffer increase plus user memory reserved must be <= 12.

LCN NODES CONFIGURATION FORM (Continued)

APPLICATION MODULE NODE DISPLAY, Continued

EXTERNAL LOAD MODULE NAMES & ASSOCIATED PERSONALITY TYPES

6

NAME

PERS

A 4x2 grid of blank coordinate planes. Each plane has a horizontal x-axis and a vertical y-axis, both represented by solid black lines. The axes intersect at the origin. The grid is composed of 8 identical coordinate planes arranged in 4 rows and 2 columns. The planes are separated by dashed vertical lines and solid horizontal lines.

```
8  ADDITIONAL MODULE MEMORY (WORDS(AMO) | | | | | | | |
```

9 FURTHER EXTERNAL DIRECTIVES? YES NO

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-7 contains instructions for Configuration Form SW88-609, Page 5. If you did not purchase custom software (External Load Modules) skip to the next section.

Table 3-7 — Instructions for Application Module Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
APPLICATION MODULE NODE DISPLAY, Continued			
6	NAME	Enter the name(s) of External Load Modules that use this node.	Get exact Load Module name from instructions with the custom software. For A ^X M values, refer to the <i>A^XM System Administration</i> manual.
8	ADDITIONAL MODULE MEMORY (WORDS)	Enter size of any additional memory to be reserved for this node.	Get this information from documentation that comes with the Load Module.
9	FURTHER EXTERNAL DIRECTIVES? YES, NO	Circle NO unless otherwise instructed by documentation with the custom software.	YES means an external Load directive file must be accessed when the node is loaded with this personality.

LCN NODES CONFIGURATION FORM (Continued)

HIWAY GATEWAY NODE DISPLAY

```

1  NODE                |  |  |
2  REDUNDANT MEMBER NODE ID  ||  |
3  HIWAY NUMBER         |  |  |

```

EXTERNAL LOAD MODULE NAMES & ASSOCIATED PERSONALITY TYPES

4	NAME	PERS
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```

6  ADDITIONAL MODULE MEMORY (WORDS)(HGO) | | | | | | | |
7  MAXIMUM EXTERNAL MODULE MEMORY (WORDS}| | | | | | | |
8  FURTHER EXTERNAL DIRECTIVES? YES      NO

```

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.8 HIWAY GATEWAY NODE CONFIGURATION FORM SW88-609

Table 3-8 contains instructions for Configuration Form SW88-609, Page 6.

Table 3-8 — Instructions for Hiway Gateway Node Configuration Form

LINE	NAME	FORM ENTRY	REMARK
HIWAY GATEWAY MODULE NODE DISPLAY			
1	NODE	Enter a number from 1 to 64.	Physical node number; refer to Table 3-4.
2	REDUNDANT MEMBER NODE ID	Enter a number from 1 to 64.	Physical node number of the optional redundant HG. This number must be greater than the primary node number.
3	HIWAY NUMBER	Enter a number from 1 to 20.	Defines a logical address for the Data Hiway connected to this HG. If the network contains an NIM, hiway numbers must be different from UCN numbers.
4	NAME	Enter the name(s) of External Load Modules that use this node.	Get exact Load Module name from instructions with the custom software.
6	ADDITIONAL MODULE MEMORY (WORDS)	Enter size of any additional memory to be reserved for this node.	Get this information from documentation that comes with the Load Module applicable personality.
7	MAXIMUM EXTERNAL MODULE MEMORY (WORDS)	Enter maximum memory words to be allowed for External Load Modules for the personality.	If this information is not in documentation that comes with the Load Module, leave blank. It can be determined at data entry time.
8	FURTHER EXTERNAL DIRECTIVES? YES, NO	Circle NO unless otherwise instructed by documentation with the custom software.	YES means an external Load directive file must be accessed when the node is loaded with this personality.

1	NODE			
2	REDUNDANT MEMBER NODE ID	<u> </u>		
3	NODE PAIR NUMBER			

4	NAME	PERS
---	------	------

A handwriting practice sheet with two columns of five rows each. Each row contains a solid top line, a dashed midline, and a solid bottom line, with vertical dashed lines dividing the space into six columns.

(HMO) | | | | | | | | | (HMI) | | | | | | | | |

(HMO) | | | | | | | | | | (HMI) | | | | | | | | | |

8 FURTHER EXTERNAL DIRECTIVES? HMO: YES NO HMI: YES NO

Network Form Instructions

3.9 HISTORY MODULE NODE CONFIGURATION FORM SW88-609

Table 3-9 contains instructions for Configuration Form SW88-609, Page 7.

Table 3-9 — Instructions for History Module Node Configuration Form

LINE	NAME	FORM ENTRY	REMARKS
HISTORY MODULE NODE DISPLAY			
1	NODE	Enter the node number to be configured.	Physical node number; refer to Table 3-4.
2	REDUNDANT MEMBER NODE ID	Enter 0	For future use/special situations, the configurator allows an entry. Normally you should enter 0.
3	NODE PAIR NUMBER	Enter a number from 1 to 20.	Assigns a logical node-pair number to this HM that is used during Volume configuration.
4	NAME	Enter the name(s) of External Load Modules that use this node.	Get exact Load Module name from instructions with the custom software.
6	ADDITIONAL MODULE MEMORY (WORDS) HMO/HMI	Enter size of any additional memory to be reserved for this node. Enter for each applicable personality.	Get this information from documentation that comes with the Load Module.
7	MAXIMUM EXTERNAL MODULE MEMORY (WORDS) HMO/HMI	Enter maximum memory words to be allowed for External Load Modules for the personality.	If this information is not in documentation that comes with the Load Module, leave blank. It can be determined at data entry time.
8	FURTHER EXTERNAL DIRECTIVES? YES, NO HMO/HMI	Circle NO unless otherwise instructed by documentation with the custom software.	YES means an external Load directive file must be accessed when the node is loaded with this personality.

[illegible]

7 FURTHER EXTERNAL DIRECTIVES? YES NO

03/2013

3.10 COMPUTING MODULE NODE CONFIGURATION FORM SW88-609

Table 3-10 contains instructions for Configuration Form SW88-609, Page 8.

Table 3-10 — Instructions for Computing Module Node Assignment Form

LINE	NAME	FORM ENTRY	REMARKS
COMPUTING MODULE NODE DISPLAY			
1	NODE	Enter the node number to be configured.	Physical node number; refer to Table 3-4.
2	ASSIGNED UNITS	Enter 1 to 62 Unit IDs in startup order (from Form SW88-506). For a CM with a redundant node, the limit is 61. Unit IDs must be separated by commas or spaces.	Defines the Units that include points assigned to this Computing Module. A Unit can be assigned to only one AM and/or one CM.
3	NAME	Enter the name(s) of External Load Modules that use this node.	Get exact Load Module name from instructions with the custom software.
5	ADDITIONAL MODULE MEMORY (WORDS)	Enter size of any additional memory to be reserved for this node.	Get this information from documentation that comes with the Load Module.
6	MAXIMUM EXTERNAL MODULE MEMORY (WORDS)	Enter maximum memory words to be allowed for External Load Modules for the personality.	If this information is not in documentation that comes with the Load Module, leave blank. It can be determined at data entry time.
7	FURTHER EXTERNAL DIRECTIVES? YES, NO	Circle NO unless otherwise instructed by documentation with the custom software.	YES means an external Load directive file must be accessed when the node is loaded with this personality.

LCN NODES CONFIGURATION FORM (Continued)

NETWORK	INTERFACE	MODULE	NODE	DISPLAY
1	2	3	4	5

1	NODE		<u> </u>		
2	REDUNDANT MEMBER NODE ID				
3	UNIVERSAL CONTROL NETWORK		<u> </u>		

EXTERNAL LOAD MODULE NAMES & ASSOCIATED PERSONALITY TYPES

<u>4</u>	NAME	PERS
----------	------	------

A handwriting practice sheet with two columns of four rows each. Each row contains a solid top line, a dashed midline, and a solid bottom line, with vertical dashed lines creating columns for letter practice.

```

6  ADDITIONAL MODULE MEMORY (WORDS) (NMO) | | | | | | | |
7  MAXIMUM EXTERNAL MODULE MEMORY (WORDS) | | | | | | | |
8  FURTHER EXTERNAL DIRECTIVES?   YES  NO

```

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.11 NETWORK INTERFACE MODULE NODE CONFIGURATION FORM SW88-609

Table 3-11 contains instructions for Configuration Form SW88-609, Page 9.

Table 3-11 — Instructions for Network Interface Module Node Configuration Form

LINE	NAME	FORM ENTRY	REMARK
NETWORK INTERFACE MODULE NODE DISPLAY			
1	NODE	Enter a number from 1 to 64.	Physical node number; refer to Table 3-4.
2	REDUNDANT MEMBER NODE ID	Enter a number from 1 to 64.	Physical node number of the optional redundant NIM. This number must be greater than the primary node number.
3	UNIVERSAL CONTROL NETWORK	Enter a number from 1 to 20.	Defines a logical address for the Universal Control Network connected to this NIM. If the system contains an HG, UCN numbers must be different from Data Hiway numbers.
4	NAME	Enter the name(s) of External Load Modules that use this node.	Get exact Load Module name from instructions with the custom software.
6	ADDITIONAL MODULE MEMORY (WORDS)	Enter size of any additional memory to be reserved for this node.	Get this information from documentation that comes with the Load Module.
7	MAXIMUM EXTERNAL MODULE MEMORY (WORDS)	Enter maximum memory words to be allowed for External Load Modules for the personality.	If this information is not in documentation that comes with the Load Module, leave blank. It can be determined at data entry time.
8	FURTHER EXTERNAL DIRECTIVES? YES, NO	Circle NO unless otherwise instructed by documentation with the custom software.	YES means an external Load directive file must be accessed when the node is loaded with this personality.

LCN NODES CONFIGURATION FORM (Continued)

NETWORK GATEWAY MODULE NODE DISPLAY

1	NODE			
2	PIN LOCAL ADDRESS			
3	PIN GLOBAL ADDRESS			
4	CONFIGURED CABLES	A	B	

EXTERNAL LOAD MODULE NAMES & ASSOCIATED PERSONALITY TYPES

5	NAME	PERS
---	------	------

7 ADDITIONAL MODULE MEMORY (WORDS) (NGO) _____

8 MAXIMUM EXTERNAL MODULE MEMORY (WORDS) _____

9 FURTHER EXTERNAL DIRECTIVES? YES **NO**

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.12 NETWORK GATEWAY MODULE NODE CONFIGURATION FORM SW88-609

Table 3-12 contains instructions for Configuration Form SW88-609, Page 10.

Table 3-12— Instructions for Network Gateway Module Node Configuration Form

LINE	NAME	FORM ENTRY	REMARK
NETWORK INTERFACE MODULE NODE DISPLAY			
1	NODE	Enter a number from 1 to 64.	Physical LCN node number; refer to Table 3-4.
2	PIN LOCAL ADDRESS	Enter a number from 1 to 64.	Enter the local logical address.
3	PIN GLOBAL ADDRESS	See Remark column.	Enter the Honeywell provided address that corresponds to the local address.
4	CONFIGURED CABLES	Select A or B or both.	Select which cables are configured.
5	NAME	Enter the name(s) of External Load Modules that use this node.	Get exact Load Module name from instructions with the custom software.
7	ADDITIONAL MODULE MEMORY (WORDS)	Enter size of any additional memory to be reserved for this node.	Get this information from documentation that comes with the Load Module.
8	MAXIMUM EXTERNAL MODULE MEMORY (WORDS)	Enter maximum memory words to be allowed for External Load Modules for the personality.	If this information isn't in documentation that comes with the Load Module, leave blank. It can be determined at data entry time.
9	FURTHER EXTERNAL DIRECTIVES? YES, NO	Circle NO unless otherwise instructed by documentation with the custom software.	YES means an external Load directive file must be accessed when the node is loaded with this personality.

NOTE

These instructions are intended to help you fill in the paper forms. The Network Data Entry manual describes a special order needed to key NG form-data into the US.

SYSTEM-WIDE VALUES CONFIGURATION

SYSTEM-WIDE VALUES MENU

SYSTEM ID DISPLAY

Page 1

1 SYSTEM DESCRIPTOR SYSTEM ID | | |

CLOCK SOURCE DISPLAY

Page 1

2 CLOCK LOCATIONS: NODE | | NODE | | |

USER AVERAGE PERIOD DISPLAY

Page 1

3 USER AVERAGE PERIOD 3 4 5 6 10 12 15 20
(Minutes)

SHIFT DATA DISPLAY

Page 1

4 SHIFTS PER WEEK | | | MONTHLY PERIOD TYPE **CALENDAR** 28 DAY
default = 21

5 REPORT START HOUR + DAY OF WEEK | | default = 0

MON TUE WED THU FRI SAT

6 REPORT START HOUR + DAY 1ST SHIFT | | default = 1

MON TUE WED THU FRI SAT

7 ARRAY OF SHIFT LENGTH HOUR - ENTER ONE VALUE FOR EACH SHIFT IN A WEEK

| | | | | | | | | | | | |
def. = 8 8 8 8 8 8 8 8 8 8 8 8

| | | | | | | | | | | | |
def. = 8 8 8 8 8 8 8 8 8 8 8

| | | | | | | | | | | | |

| | | | | | | | | | | | |

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.13 SYSTEM-WIDE VALUES CONFIGURATION FORM SW88-615

Table 3-13 contains instructions for Configuration Form SW88-615.

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 1 of 12)

LINE	NAME	FORM ENTRY	REMARKS
SYSTEM-WIDE VALUES MENU DISPLAY			
1	SYSTEM DESCRIPTOR	Enter up to 24 characters.	System descriptor, which consists of any 24 characters, is user-defined.
	SYSTEM ID	Enter three alphanumeric characters.	
2	CLOCK NODE 1	Enter a number from 1 to 64 (lowest number in left box).	The system must have two clock source paddle boards. If any nodes have HMPU or HPK2 boards, two of those with clock paddle boards must be used as clock sources. CLOCK NODE 1 and CLOCK NODE 2 are the Master and Slave clock sources. The clock node that is loaded first with its personality is designated as the Master clock source. The remaining node is designated as the Slave clock source. If the Master clock source fails, then the Slave clock source becomes the Master.
	CLOCK NODE 2	Enter a number from 1 to 64 (highest number in right box).	
NOTE With GUS R330 and APP R211 or later, the TPN Time Sync function provides the ability to synchronize the TPN time to the TPS node time. This function is not applicable to the TPS Universal GUS nodes. If you are implementing the TPN Time Sync function, CLOCK NODE 1 and CLOCK NODE 2 clock source nodes must have LCNP4-GPS boards (or later) installed. Assign CLOCK NODE 1 and CLOCK NODE 2 to the appropriate GUS and/or APP node numbers when implementing the TPN Time Sync function. In addition, all TPN nodes must have the K2LCN board (or later) installed. With TPN R682, the Clock Master and Slave nodes can be switched online and does not require reboot of any node in the network. During the Clock Mastership transition, the nodes shall go to Local mode. After the Master / Slave transition is complete all the nodes become Listener to the new Master. This option is available in all the LCN node personalities. This feature is limited to the nodes with K2LCN, K4LCN boards and LCNP4 boards that support digital clocks. However, this feature does not support the HMPU, HPK2, EMPU and MCPU boards which support analog clocks.			

(Continued)

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 1 of 12)
(Continued)

LINE	NAME	FORM ENTRY	REMARKS
SYSTEM-WIDE VALUES MENU DISPLAY			
3	USER AVERAGE PERIOD (Minutes)	Circle one: 3 4 5 6 10 12 15 20 30	Selects the period for Continuous History user average groups on all HMs.
4	SHIFTS PER WEEK	Enter a number from 1 to 42. (Default for 8-hour shifts is 21.)	Sets the total number of shifts that occur in a week.
	MONTHLY PERIOD TYPE	Circle one: CALENDAR 28 DAY	Selects the base period for a month of Continuous History.
5	REPORT START HOUR AND DAY OF WEEK	Enter 1 or 2 numeric digits Circle one: MON TUE WED THU FRI SAT SUN	Defines the day and hour when the week begins. If shift begins at midnight, enter 0.
6	REPORT START HOUR AND DAY OF FIRST SHIFT	Enter 1 or 2 numeric digits Circle one: MON TUE WED THU FRI SAT SUN	Defines the day and hour when the first shift begins. If shift begins at midnight, enter 0.
7	ARRAY OF SHIFT LENGTH HOUR – ENTER ONE VALUE FOR EACH SHIFT IN A WEEK	Enter a number of nonzero fields equal to the number of shifts per week defined on line 6. For 21, 8-hour shifts, enter the number eight, followed by space or comma, 21 times.	Each 1- or 2-digit number represents shift length in hours.

SYSTEM-WIDE VALUES CONFIGURATION FORM
(Continued)

CONSOLE DATA DISPLAY										PAGE 1																																																																															
NCF INSTALL TIME										DATE/TIME										FILE VERSION										//										REVISION										//																																							
NCF SOURCE										PATHNAME										CONFIGURATOR										//																				//																																							
8										FAST RAISE/LOWER RATE										2										3										5										10										PERCENT PER INCREMENT																													
9										PRINTER PAGE LENGTH																																								PRINTER PAGE WIDTH																																							
																				default = 66																																								default = 132																													
10										TREND PEN DELAY																				1 SEC																				6 SEC																																							
11										CL MESSAGE CONFIRMATION-AFTER-ACKNOWLEDGEMENT										YES																				NO																																																	
12										PRINTED TREND FORMAT																				GRAPHIC																				NUMERIC																																							
13										REAL TIME JOURNAL SIZE																				100																				400																																							
14										REAL TIME JOUR. HEADER/TRAILER MESSAGES																				BEG/END																				BEG ONLY																																							
16										ACTOR KEYLEVEL CHANGE JOURNAL SUPPRESSION																				YES																				NO																																							
17										CONTACT CUTOUT ALARMS ON ALARM SUMMARY										CLEAR IMMEDIATELY																				CLEAR WHEN ACKED																																																	
CONSOLE DATA DISPLAY																				PAGE 2																																																																					
HORN ANNUNCIATION POLICY																																																																																									
18										CONSOLE STATUS EVENT										CONTACT_1										CONTACT 2										CONTACT 3																																																	
19										SYSTEM STATUS EVENT										CONTACT_1										CONTACT 2										CONTACT 3																																																	
20										OPERATOR ACK ONLY MESSAGE										CONTACT_1										CONTACT 2										CONTACT 3																																																	
21										OPERATOR CONFIRM MESSAGE										CONTACT 1										CONTACT_2										CONTACT 3																																																	
22										LOW PRIORITY ALARM										CONTACT_1										CONTACT 2										CONTACT 3																																																	
23										HIGH PRIORITY ALARM										CONTACT 1										CONTACT_2										CONTACT 3																																																	
24										EMERGENCY PRIORITY ALARM										CONTACT 1										CONTACT 2										CONTACT_3																																																	
25										CONTACT 1 OUTPUT STATE										MOMENTARY										STEADY																																																											
26										CONTACT 2 OUTPUT STATE										MOMENTARY										STEADY																																																											
27										CONTACT 3 OUTPUT STATE										MOMENTARY										STEADY																																																											
28										PRINTER ALARMS ANNUNCIATED										YES										NO																																																											
References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder. Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.																																																																																									

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 1 of 12)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 1			
	NCF RELATED DATA	No entry required	For reference only.
8	FAST RAISE/LOWER RATE (PERCENT PER INCREMENT)	Circle one: 2 3 5 10	Sets the raise/lower rate for the double-arrow increment buttons on the operator's keyboard.
9	PRINTER PAGE—LENGTH	Enter a number from 0 to 255 (Default = 66)	Defines the number of lines for each printed page.
	WIDTH	Enter a number from 80 to 132. (Default = 132)	For the date to be printed on every page, the entered value should not exceed the actual page length . Defines the maximum width (in character units) of a printed page).
10	TREND PEN DELAY	Circle one 1 SEC or 6 SEC	Select the delay period for marking (moving the pen from 0 to 100%).
11	CL MESSAGE CONFIRMATION AFTER ACKNOWLEDGEMENT	Circle one: YES NO	Select YES to allow message confirmation only after acknowledgement. Select NO to allow confirm before ACK.
12	PRINTED TREND FORMAT	Circle one: GRAPHIC or NUMERIC	Select Graphic for traditional format or Numeric for numerical format.
13	REAL TIME JOURNAL JOURNAL SIZE	Circle one: 100 or 400 (Default = 100)	Defines the RTJ Event Buffer Size (No. of Events).
14	REAL TIME JOUR. HEADER/TRAILER MESSAGE	Circle one: BEG/END or BEG ONLY	Select BEG ONLY to print the RTJ header with the date and time of the immediately succeeding RTJ event. Select BEG/END to print the RTJ header with the date and the time of the immediately succeeding RTJ event and the RTJ trailer with the date of the immediately preceding RTJ event.
16	ACTOR KEYLEVEL CHANGE JOURNAL SUPPRESSION	Circle one: YES NO (R520 and later software)	YES = suppress HM/RTJ journaling of keylevel changes due to \$KEYCHG, \$KEYRST, and CROSSKBD actors.

(Continued)

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 1 of 12) (Continued)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 1			
17	CONTACT CUTOFF ALARMS ON ALARM SUMMARY DISPLAY	Circle one: CLEAR IMMEDIATELY CLEAR WHEN ACKED	Choose how unacknowledged Contact Cutoff alarms are cleared. If set to Clear when ACKED, alarms remain on the Alarm Summary display with the timestamp backlit until Acknowledged. Defines relay contact to be activated at this US when:

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 2 of 12) (Continued)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM -WIDE VALUES DISPLAY—PAGE 2			
18	CONSOLE STATUS EVENT	Circle 1 to 3: CONTACT 1 CONTACT 2 CONTACT 3	—a console status alarm or console status change occurs.
19	SYSTEM STATUS EVENT	Circle 1 to 3: CONTACT 1 CONTACT 2 CONTACT 3	—a system alarm or system status change occurs.
20	OPERATOR ACK ONLY MESSAGE	Circle 1 to 3: CONTACT 1 CONTACT 2 CONTACT 3	—an operator acknowledges a message at the Message Summary display.
21	OPERATOR CONFIRM MESSAGE	Circle 1 to 3: CONTACT 1 CONTACT 2 CONTACT 3	—an operator confirms a message at the Message Summary display.
22	LOW PRIORITY ALARM	Circle 1 to 3: CONTACT 1 CONTACT 2 CONTACT 3	—a low priority process alarm occurs.
23	HIGH PRIORITY ALARM	Circle 1 to 3: CONTACT 1 CONTACT 2 CONTACT 3	—a high priority process alarm occurs.
24	EMERGENCY PRIORITY ALARM	Circle 1 to 3: CONTACT 1 CONTACT 2 CONTACT 3	—an emergency priority process alarm occurs.
25	CONTACT 1 OUTPUT STATE	Circle one: Momentary or Steady	—Horn sounds for two seconds or horn stays on (until reset).
26	CONTACT 2 OUTPUT STATE	Circle one: Momentary or Steady	—Horn sounds for two seconds or horn stays on (until reset).
27	CONTACT 3 OUTPUT STATE	Circle one: Momentary or Steady	—Horn sounds for two seconds or horn stays on (until reset).
28	PRINTER ALARMS ANNUNCIATED	Circle one: Yes or No	—Yes = Printer alarms on the console will sound and Console Status key LED will light. No = journal and show Service Alarm (no blink) on Console Status display (but no sound).

SYSTEM-WIDE VALUES CONFIGURATION FORM
(Continued)

CONSOLE DATA DISPLAY										PAGE 3	
ALARM MANAGEMENT CONFIGURATION											
29	RED COLOR ALARM PRIORITY	LOW	HIGH	EMERGENCY							
30	PRIORITY INDICATORS ON ALARM DISPLAYS	CHARACTERS		SYMBOLS							
31	ALARM PRIORITY COLORS	RED, YELLOW	3 COLORS (USER SELECTABLE)								
32	THREE COLOR ALARM PRIORITY:										
	EMERGENCY	RED	YELLOW	MAGENTA	GREEN	BLUE	CYAN	WHITE			
	HIGH	RED	YELLOW	MAGENTA	GREEN	BLUE	CYAN	WHITE			
	LOW	RED	YELLOW	MAGENTA	GREEN	BLUE	CYAN	WHITE			
33	LOW PRIORITY ALARMS IN AREA ALARM SUMMARY	YES	NO								
34	INITIAL ALARM SUMMARY SORT STATE	CHRONOLOGICAL		PRIORITY							
35	AUDIBLE ALARM ANNUNCIATION SUPPRESSION TIME OUT (SECONDS) default = 0										
36	ALARM SUMMARY FREEZE TIME-OUT (SECONDS) default = 0										
37	AREA-BASED PROCESS ALARM ACKNOWLEDGE?	YES	NO								
38	CONSOLE-WIDE SILENCE BUTTON? (R520 and later)	YES	NO								

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 3 of 12)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 3			
29	RED COLOR ALARM PRIORITY	Circle one: LOW HIGH EMERGENCY	Select the alarm that has the lowest priority that will be associated with the color red. If 3-color option (below) is selected, it applies only to LEDs on the Alarm Summary key.
30	PRIORITY INDICATORS ON ALARM DISPLAYS	Circle one: CHARACTERS or SYMBOLS	Characters = E, H, L. Symbols = ☐ <input type="triangle-up"/> <input type="triangle-down"/> for Emerg, High, Low alarms.
31	ALARM PRIORITY COLORS	Circle one: Red, Yellow or 3 Colors (User Selectable)	Alarm Priority Colors: 2-color (configurable) or 3-color, select below.
32	EMERGENCY HIGH LOW	Circle choice for each level: Red, Yellow, Magenta, Green, Blue, Cyan, White	Selectable Alarm Priority Colors. If 3-Color Option (above) is selected, choose one color for each alarm level.
33	LOW PRIORITY ALARMS IN AREA ALARM SUMMARY	Circle one: YES or NO	Select Yes to add low Priority Alarms to the Area Alarm Summary.
34	INITIAL ALARM SUMMARY SORT STATE	Circle one: Chronological or Priority	Display alarms in reverse <u>chronological</u> order or first sort by descending alarm <u>priority</u> , then chronologically reverse within each alarm priority level.
35	AUDIBLE ALARM ANNUNCIATION SUPPRESSION TIME OUT	Enter a number from 0 to 32767	Enter the period in seconds that Audible Alarms are suppressed. WARNING: Do not configure an overly long timeout.
36	ALARM SUMMARY FREEZE TIME-OUT	Enter a number from 0 to 32767	Enter the period in seconds after which alarm scrolling automatically resumes. WARNING: Do not configure an overly long timeout.
37	AREA-BASED PROCESS ALARM ACKNOWLEDGEMENT	Circle one: YES or NO	Yes = enable Area based process alarm acknowledgement.
38	CONSOLE-WIDE SILENCE BUTTON	Circle one: YES or NO (R520 and later)	Yes = Pressing the silence button at any US in a console turns off all horns in that console. No = the SIL button turns off horns only at stations in a console in the same area as the alarm(s).

SYSTEM-WIDE VALUES CONFIGURATION FORM
(Continued)

CONSOLE DATA RTJ PRINT LINE COLOR DISPLAY							PAGE 4
38a	ALARM SUMMARY	PARAMETER FOR OP ALARM (R670 and later)	PV				OP
38b	ALENBST	INDICATION IN JOURNAL FOR DISABLED ALARMS	YES				NO
39	EMERGENCY PRIORITY:	(R520 and later)					
	BLACK	RED	YELLOW	CYAN	ORANGE	GREEN	PURPLE BROWN
	HIGH PRIORITY						
	BLACK	RED	YELLOW	CYAN	ORANGE	GREEN	PURPLE BROWN
	LOW PRIORITY						
	BLACK	RED	YELLOW	CYAN	ORANGE	GREEN	PURPLE BROWN
	PRINT ONLY						
	BLACK	RED	YELLOW	CYAN	ORANGE	GREEN	PURPLE BROWN
	JOURNAL PRINT						
	BLACK	RED	YELLOW	CYAN	ORANGE	GREEN	PURPLE BROWN
	RETURN TO NORMAL						
	BLACK	RED	YELLOW	CYAN	ORANGE	GREEN	PURPLE BROWN
39a	PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN		YES				NO

CONSOLE DATA DISPLAY							PAGE 5
	LCN CABLE SWAP INHIBIT						
40	LCN CABLE SWAP INHIBIT TIME IN MINUTES						
	(R510 and later)						
41	ACCESS LEVEL TO INHIBIT LCN CABLE SWAPPING	OPER	SUPER	ENGR			
	(R510 and later)						
	NCF ACCESS						
	ACCESS LEVEL TO VIEW NCF (R620 and later)	OPER	SUPER	ENGR			
	ALARM COMMAND ACCESS TABLE						
42	CHANGE ALARM SUMMARY SORT STATE	OPER	SUPER	ENGR			
43	CHG ALARM PRIORITY FILTER ON ALARM SUMMARIES	OPER	SUPER	ENGR			
44	CHG AUDIBLE ALARM ANNUNCIATION SUPPRESSION	OPER	SUPER	ENGR			
45	ALARM SUMMARY DISPLAY FREEZE	OPER	SUPER	ENGR			
46	AUXILIARY UNIT CHANGE (R520 and later)	OPER	SUPER	ENGR			
	GROUP/DETAIL DISPLAY						
47	GROUP DETAIL DISPLAY (R520 and later)						
	PARAMETER CHANGES/CONTROL ACTIONS	OPER	SUPER	ENGR			

Table 3-13 — Instructions for System-Wide Values Configuration Form (Pages 4 and 5 of 12)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 4			
38a	ALARM SUMMARY PARAMETER FOR OP ALARM	Select PV or OP	PV = PV value is displayed for all the output alarms in the Alarm Summary Display. OP = OP value is displayed for all the output alarms in the Alarm Summary Display.
38b	ALENBST INDICATION IN JOURNAL FOR DISABLED ALARMS	Circle one: YES NO (R680 and later software)	YES = indicates ALENBST state for disabled alarms in Process Alarm journal entry.
39	RTJ PRINTER LINE COLOR: Emergency, High/Low Priority, Print/Journal Print and Return To Normal	Circle a choice for each RTJ alarm: Black, Red, Yellow, Cyan, Orange, Green, Purple, or Brown (R520 and later)	The Real Time Journal printer must be a Honeywell supported ASPI printer with a color ribbon to print colors other than black.
39a	PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN	Circle one: YES NO (R683 and later software)	YES = displays and prints the NIM priority parameters and the related parameters when the trip points are set to NaN in PED and .EB files respectively.
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 5			
40	LCN CABLE SWAP INHIBIT TIME, MINUTES	Enter a number from 0 to 9999 (R510 and Later)	Enter the period in MINUTES that the LCN cables will be inhibited from switching.
41	ACCESS LEVEL TO INHIBIT LCN CABLE SWAPPING ACCESS LEVEL TO VIEW NCF	Circle one: OPER SUPER ENGR (R510 and Later) Circle one: OPER SUPER ENGR (R620 and Later)	- Disable LCN Cable swapping for the period configured above. - Access level to view NCF displays.

(Continued)

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 5 of 12) (Continued)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 5			
42	CHANGE ALARM SUMMARY SORT STATE	Alarm Commands Access Circle one: OPER SUPER ENGR	Specify the lowest keylock access level <u>needed</u> to: - Change Alarm Summary Sort State from chronological to priority sorting.
43	CHANGE ALARM PRIORITY FILTER ON ALARM SUMMARIES	Circle one: OPER SUPER ENGR	- Change Alarm Priority Filter on Area and Unit Alarm Summaries, for example to filter out all but Emergency alarms.
44	CHANGE AUDIBLE ALARM ANNUNCIATION SUMMARIES	Circle one: OPER SUPER ENGR	- Suppress Audible Alarm Annunciation by using the AUDIBLE target on the Alarm Summary and Alarm Annunciation displays.
45	ALARM SUMMARY DISPLAY FREEZE	Circle one: OPER SUPER ENGR	- Freeze the Alarm Summary display. If = 0, no target appears on alarm displays.
46	AUXILIARY UNIT CHANGE	Circle one: OPER SUPER ENGR (R520 and later)	- Modify/add the \$AUXUNIT parameter to a given point from the Detail display or from a custom schematic.
47	GROUP/DETAIL DISPLAY—PARAMETER CHANGES / CONTROL ACTIONS	Circle one: OPER SUPER ENGR (R520 and later)	- Permit changes to standard changeable parameters and control actions through the Group or Point Detail displays. Takes precedence over other display keylevels.

SYSTEM-WIDE VALUES CONFIGURATION FORM
(Continued)

CONSOLE DATA DISPLAY				PAGE 6	
CONFIGURABLE ACCESS LEVEL ON PARAMETERS (R530)					
48	ALARM LIMITS	DEFAULT	OPER	SUPER	ENGR
49	CONTROL LIMITS	DEFAULT	OPER	SUPER	ENGR
50	RANGE LIMITS	DEFAULT	OPER	SUPER	ENGR
51	TUNING PARAMETERS	DEFAULT	OPER	SUPER	ENGR
PHYSICAL NODE COMMAND ACCESS					
52	LOAD OPERATIONS NON US NODE (CG/NG/AM ONLY)	OPER	SUPER	ENGR	
53	LOAD OPERATIONS LOCAL CONSOLE	OPER	SUPER	ENGR	
54	LOAD OPERATIONS REMOTE CONSOLE	OPER	SUPER	ENGR	
55	DUMP NODE	OPER	SUPER	ENGR	
56	OVERRIDE DUMP	OPER	SUPER	ENGR	
57	SHUTDOWN NODE	OPER	SUPER	ENGR	
58	OVERRIDE SHUTDOWN	OPER	SUPER	ENGR	
CONSOLE DATA DISPLAY				PAGE 7	
PHYSICAL NODE COMMAND ACCESS					
59	ENABLE/DISABLE LOAD/DUMP	OPER	SUPER	ENGR	
60	CHANGE TIME OR DATE	OPER	SUPER	ENGR	
61	CHANGE AREA	OPER	SUPER	ENGR	
62	OVERRIDE CHANGE AREA	OPER	SUPER	ENGR	
63	CHANGE ALENBST	OPER	SUPER	ENGR	
64	IDLE PROCESS MANAGER/PMM/IOP	OPER	SUPER	ENGR	
65	DEMAND CHECKPOINT NODE	OPER	SUPER	ENGR	
66	ENABLE DISABLE AUTO CHECKPOINT	OPER	SUPER	ENGR	
67	SWITCH PRIMARY TO SECONDARY	OPER	SUPER	ENGR	
68	OVERRIDE SWITCH PRIMARY TO SECONDARY	OPER	SUPER	ENGR	
69	ENABLE/DISABLE HISTORY COLLECTION	OPER	SUPER	ENGR	

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 6 of 12)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 6			
		Circle one each (R530 or later):	
48	ALARM LIMITS	DEFAULT OPER SUPER ENGR	The access levels will override to access level for the parameters contained in the Parameter Reference Dictionary. It will be overwritten by the Group/Detail display parameter change setting if the Group/Detail is more restrictive.
49	CONTROL LIMITS	DEFAULT OPER SUPER ENGR	
50	RANGE LIMITS	DEFAULT OPER SUPER ENGR	
51	TUNING PARAMETERS	DEFAULT OPER SUPER ENGR	
52	LOAD OPERATIONS NON-US NODE (CG/NG/AM only)	Circle one: OPER SUPER ENGR	–load non-US nodes from any US.
53	LOAD OPERATIONS LOCAL CONSOLE	Circle one: OPER SUPER ENGR	–load a US node in the local console.
54	US	Circle one: OPER SUPER ENGR	–load a US node in the remote console.
55	LOAD OPERATIONS REMOTE CONSOLE US	Circle one: OPER SUPER ENGR	–perform a dump-memory operation on a node.
56	DUMP NODE	Circle one: OPER SUPER ENGR	–override a dump-node request.
57	OVERRIDE DUMP	Circle one: OPER SUPER ENGR	–shut down a node.
58	SHUTDOWN NODE	Circle one: OPER SUPER ENGR	–override a node-shutdown request. Override is necessary to shut down a node performing a primary function such as when it is accessing configuration screens.
	OVERRIDE SHUTDOWN		

(Continued)

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 7 of 12) (Continued)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 7			
59	ENABLE/DISABLE LOAD/DUMP	Circle one: OPER SUPER ENGR	—enable or disable load or dump operations for any node.
60	CHANGE TIME OR DATE	Circle one: OPER SUPER ENGR	—change the system time or date. Defines lowest keylock access level needed to:
61	CHANGE AREA	Circle one: OPER SUPER ENGR	—perform an area database- change operation on any US, or in local console.
62	OVERRIDE CHANGE AREA	Circle one: OPER SUPER ENGR	—override an area database- change request.
63	CHANGE ALENBST	Circle one: OPER SUPER ENGR	—enable/inhibit point Alarm Enable Status.
64	IDLE PROCESS MANAGER/PMM/IOP	Circle one: OPER SUPER ENGR	—idle PM/APM/HPM from UCN Status display or idle <u>any</u> modules on PM/ APM/HPM Detail Status displays. See also Process Network Fast Commands (display page 8).
65	DEMAND CHECKPOINT NODE	Circle one: OPER SUPER ENGR	—request an immediate check- pointing of control data from HG, AM, and so on.
66	ENABLE/DISABLE AUTO CHECKPOINT	Circle one: OPER SUPER ENGR	—enable or disable automatic checkpointing at an HG, AM, and so on.
67	SWITCH PRIMARY TO SECONDARY	Circle one: OPER SUPER ENGR	—not implemented.
68	OVERRIDE SWITCH PRIMARY TO SECONDARY	Circle one: OPER SUPER ENGR	—not implemented.
69	ENABLE/DISABLE HISTORY COLLECTION	Circle one: OPER SUPER ENGR	—enable or disable history collection at an HM.

SYSTEM-WIDE VALUES CONFIGURATION FORM
(Continued)

CONSOLE DATA DISPLAY PAGE 8

PHYSICAL NODE COMMAND ACCESS

70	HOT/WARM/COLD/NO PROCESS START AM	OPER	SUPER	ENGR
71	SPARE	OPER	SUPER	ENGR
72	PRINTER ASSIGNMENT CHANGE	OPER	SUPER	ENGR
73	CLEAR MAINTENANCE	OPER	SUPER	ENGR
74	LOAD TEST	OPER	SUPER	ENGR
75	LOAD INIT	OPER	SUPER	ENGR
76	NO POINT PROCESS LOAD	OPER	SUPER	ENGR
77	COMMAND LOAD OF PROCESS GATEWAY NODE (HG AND NIM)	OPER	SUPER	ENGR
78	COMMAND RESTORE OF PROCESS NETWORK	OPER	SUPER	ENGR
79	COMMAND LOAD OF PROCESS NETWORK	OPER	SUPER	ENGR
80	COMMAND CHECKPT OF PROCESS NETWORK	OPER	SUPER	ENGR

CONSOLE DATA DISPLAY PAGE 9

MISCELLANEOUS COMMANDS ACCESS

81	SEQUENCE COMMAND LOAD PROGRAM	OPER	SUPER	ENGR
82	SEQUENCE COMMAND SAVE PROGRAM	OPER	SUPER	ENGR
83	BOX VARIABLE CHANGE ACCESS	OPER	SUPER	ENGR
84	SYSTEM EVENT ACKNOWLEDGE ACCESS	OPER	SUPER	ENGR
85	FLOPPY INIT ACCESS	OPER	SUPER	ENGR
86	ACCESS LEVEL TO CHANGE VIEW ACCESS LEVEL	OPER	SUPER	ENGR
87	ACCESS LEVEL TO CHANGE UNIT ASSIGNMENT	OPER	SUPER	ENGR
88	ACCESS LEVEL TO CHANGE UNIT DEASSIGNMENT	OPER	SUPER	ENGR
89	ACCESS LEVEL TO CHANGE UNIT ALARM ENABLE	OPER	SUPER	ENGR
90	ACCESS LEVEL TO CHANGE UNIT ALARM DISABLE	OPER	SUPER	ENGR
91	ACCESS LEVEL TO CHANGE UNIT ALARM INHIBIT	OPER	SUPER	ENGR
92	PROCESS NETWORK FAST COMMANDS	OPER	SUPER	ENGR

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 8 of 12)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 8			
70	HOT/WARM/COLD/ NO PROCESS START AM	Circle one: OPER SUPER ENGR	–restart an AM.
71	Reserved For Future Use		Defines lowest keylock access level needed to:
72	PRINTER ASSIGNMENT CHANGE	Circle one: OPER SUPER ENGR	–change the printer assignment at a Universal Station.
73	CLEAR MAINTENANCE	Circle one: OPER SUPER ENGR	–clear the maintenance display at a Node Status or Console Status display.
74	LOAD TEST	Circle one: OPER SUPER ENGR	–load a test personality into a module.
75	LOAD INIT	Circle one: OPER SUPER ENGR	–load an Initialization personality into an HM.
76	NO POINT PROCESS LOAD	Circle one: OPER SUPER ENGR	–AM only.
77	COMMAND LOAD OF PROCESS GATEWAY NODE (HG and NIM)	Circle one: OPER SUPER ENGR	–request loading of the process database into HGs/ NIMs, followed by restart.
78	COMMAND RESTORE OF PROCESS NETWORK DEVICES	Circle one: OPER SUPER ENGR	–request restore of the Data Hiway Boxes/ Process Manager IOMs.
79	COMMAND LOAD OF PROCESS NETWORK DEVICES	Circle one: OPER SUPER ENGR	–request loading of the Data Hiway Boxes/Process Mgr IOPs followed by restart.
80	COMMAND CHECKPT OF PROCESS NETWORK DEVICES	Circle one: OPER SUPER ENGR	–request checkpoint of the Data Hiway Boxes and or Process Manager IOMs.

(Continued)

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 9 of 12) (Continued)

LINE	NAME	FORM ENTRY	REMARKS
CONSOLE DATA—SYSTEM-WIDE VALUES DISPLAY—PAGE 9			
81	SEQUENCE COMMAND LOAD PROGRAM	Circle one: OPER SUPER ENGR	Defines lowest keylock access level needed to: –load sequence programs.
82	SEQUENCE COMMAND SAVE PROGRAM	Circle one: OPER SUPER ENGR	–save sequence programs.
83	BOX VARIABLE CHANGE ACCESS	Circle one: OPER SUPER ENGR	–make changes at the Box Variable display.
84	SYSTEM EVENT ACKNOWLEDGE ACCESS	Circle one: OPER SUPER ENGR	–acknowledge system events at a Universal Station.
85	FLOPPY INIT ACCESS	Circle one: OPER SUPER ENGR	–use the US floppy-disk Initialize display functions.
86	ACCESS LEVEL TO CHANGE VIEW ACCESS LEVEL	Circle one OPER SUPER ENGR	–access level required to change access level associated with OPERATOR keylock position.
NOTE Lines 89-91 (ACCESS LEVEL TO CHANGE UNIT ALARM ENABLE/DISABLE/INHIBIT keylocks) determine who can change the alarm enable/disable/inhibit for each unit on a system-wide basis (SYSTEM ALARM STATE). Lines 87 and 88 (ACCESS LEVEL TO CHANGE UNIT ASSIGNMENT/DEASSIGNMENT keylocks) determine who can assign/deassign units on a console basis (CONSOLE ALARM STATE). This is basically the same as enabling/disabling alarms for a particular unit on a console.			
87	ACCESS LEVEL TO CHANGE UNIT ASSIGNMENT	Circle one: OPER SUPER ENGR	–add control and alarm capability for points in this unit to a console.
88	ACCESS LEVEL TO CHANGE UNIT DEASSIGNMENT	Circle one: OPER SUPER ENGR	–remove control and alarm capability for points in this from a console unit (see above).
89	ACCESS LEVEL TO CHANGE UNIT ALARM ENABLE	Circle one: OPER SUPER ENGR	–add alarm annunciation for points in this unit to all US's in system. No effect on HM or RT journal.
90	ACCESS LEVEL TO CHANGE UNIT ALARM DISABLE	Circle one: OPER SUPER ENGR	–remove alarm annunciation for points in this unit from all US's in system (see above).
91	ACCESS LEVEL TO CHANGE UNIT ALARM INHIBIT	Circle one: OPER SUPER ENGR	–remove all alarm annunciation and journaling for points in this unit on a system-wide basis. Stops HM & RT Jnl alarms, US annuc.
92	PROCESS NETWORK FAST COMMANDS	Circle one: OPER SUPER ENGR	–use the Fast Commands (see below).
Fast Hiway cmds: Swap Hiway, Init Addr. Fast Box cmds: Start, Idle, Enable Proc, ACK, Config Clear, Start RC, Start/Disable/Enable/Test/Swback/Save Pri.			

SYSTEM-WIDE VALUES CONFIGURATION FORM
(Continued)

HONEYWELL RESERVED SYSTEM VALUES

PAGES 10

WARNING: DO NOT CHANGE THIS PAGE UNLESS REQUESTED TO DO SO BY HONEYWELL

	BOARD ID	MINIMUM BOARD REV	FIRMWARE TYPE	FIRMWARE VERS .	MINIMUM FIRMWARE REV
110	HMPU	A	1	0	A
111	HPK2-2MW	A	1	0	A
	HPK2-3MW	A	1	0	A
112	K2LCN-2MW	B	1	0	A
	K2LCN-3MW	B	1	0	A
	K2LCN-4MW	B	1	0	A
	K2LCN-6MW	B	1	0	A
	K2LCN-8MW	B	1	0	A
	K4LCN	C	1	0	A
	K6LCN	A	1	0	A
113	KERNEL	A	1	0	A
114	MEM 256KW	D			
	MEM 256KW	C			
	MEM 512KW	C			
	MEM 768KW	C			
	MEM 1024KW	C			
	PMEM 2048KW	A			
	EMEM 1024KW	E			

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 10 of 12)

LINE	NAME	FORM ENTRY	REMARKS
RESERVED SYSTEM VALUES—SYSTEM -WIDE VALUES DISPLAY—PAGE 10			
		None	
110	HMPU	If instructed by Honeywell, board and/or firmware revision levels may need to be updated.	Required values are supplied with each software release.
111	HPK2		
112	K2LCN K4LCN K6LCN		
113	KERNEL		
114	MEM PMEM EMEM	The form shown is typical and may not exactly match the screen display for the current software release.	CAUTION After the system has been configured, changing the revision records in System-Wide Values may require shutting down/starting up one or more nodes on the LCN. Do not make these changes unless requested by Honeywell

SYSTEM-WIDE VALUES CONFIGURATION FORM
(Continued)

HONEYWELL RESERVED SYSTEM VALUES DISPLAY

PAGES 11

WARNING: DO NOT CHANGE THIS PAGE UNLESS REQUESTED TO DO SO BY HONEYWELL

	BOARD ID	MINIMUM BOARD REV	FIRMWARE TYPE	FIRMWARE VERS .	MINIMUM FIRMWARE REV
	QMEM 2 048 KW	A			
	QMEM 3 072 KW	A			
	QMEM 4 096 KW	A			
125	SIO	B			
126	AMR	E			
	EAMR	A			
127	APDL	A			
129	WSI2	A			
130	VDG	D			
131	PIC	H	1	1	H
132	PDG	B	1	1	B
133	EPDG	D	1	1	C
	TPDG	A	1	1	A
134	LCNI	F	1	0	C
135	LCNI -2	A	1	0	A
	LCNI -4	A	1	0	A
138	CLMI -T	A	1	5	A

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 11 of 12)

LINE	NAME	FORM ENTRY	REMARKS
RESERVED SYSTEM VALUES—SYSTEM -WIDE VALUES DISPLAY—PAGE 11			
	QMEM2048KW QMEM3072KW QMEM4096KW		
125	SIO		
126	AMR		
127	EAMR		
	APDL		
129	WSI2		
130	VDG		
131	PIC		
132	PDG		
133	EPDG		
	TPDG		
134	LCNI		
135	LCNI-2		
	LCNI-4		
138	CLMI-T		

SYSTEM-WIDE VALUES CONFIGURATION FORM
(Continued)

HONEYWELL RESERVED SYSTEM VALUES DISPLAY

PAGES 12

WARNING: DO NOT CHANGE THIS PAGE UNLESS REQUESTED TO DO SO BY HONEYWELL

	BOARD ID	MINIMUM BOARD REV	FIRMWARE TYPE	FIRMWARE VERS .	MINIMUM FIRMWARE REV
139	CLI	E	1	1	D
140	CNI	C	1	1	C
141	PNI	A	1	0	E
142	EPNI	A	1	0	B
143	NGI	A	1	0	A
145	DHI	E	1	0	J
146	PLCI	B	1	0	J
147	AMGI	A	2	0	A
148	WDC	L	1	0	G
149	SPCW	B	1	0	B
150	FDC	G	1	0	J
151	SPCF	B	1	0	B
152	SCSI EPDG	A	1	1	A
154	SCSI TPDG	A	1	1	A
155	EPNI2	A	1	0	A

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 12 of 12)

LINE	NAME	FORM ENTRY	REMARKS
RESERVED SYSTEM VALUES—SYSTEM -WIDE VALUES DISPLAY—PAGE -12			
139	CLI		
140	CNI		
141	PNI		
142	EPNI		
143	NGI		
145	DHI		
146	PLCI		
147	AMGI		
148	WDC		
149	SPCW		
150	FDC		
151	SPCF		
152	SCSI_EPDG		
154	SCSI_TPDG		
155	EPNI2		

NOTE

Software Options—Purchased software options are entered through the System Wide Values display, but no configuration form is necessary because the options are read directly from a floppy diskette or cartridge disk supplied by Honeywell. Note that Software Options are not the same as the Custom Software Load Modules mentioned in LCN Node Configuration.

For additional information on Software Options, refer to the *Engineer's Reference Manual*.

SYSTEM-WIDE VALUES CONFIGURATION FORM

NETWORK GATEWAY (NG)		LOCAL SYSTEM ID		Page 1	
160	LOCAL SYSTEM NUMBER				default = 0
161	SYSTEM TITLE				
162	SYSTEM DESCRIPTION				
		_	_	_	_
		_	_	_	_
		_	_	_	_
		_	_	_	_
163	PIN ADDRESSING FORM:	LOCAL		GLOBAL	
164	GLOBAL VENDOR NUMBER:				

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form — NG LOCAL SYSTEM ID

LINE	NAME	FORM ENTRY	REMARKS
NETWORK GATEWAY—LOCAL SYSTEM ID DISPLAY			
160	LOCAL SYSTEM NUMBER	Enter a number from 1 to 64.	Uniquely identifies the (local) LCN among the systems interconnected through the Plant Information Network (PIN). The local LCN is the LCN connected to the US you are using.
161	SYSTEM TITLE	Enter two characters. The first must be an alpha.	Two-character identifier for the local LCN number.
162	SYSTEM DESCRIPTION	Enter up to 20 characters.	Descriptor for the local LCN.
163	PIN ADDRESSING FORM	Circle one: LOCAL GLOBAL	Select Local or Global. If you select Global, enter the global vendor number below.
164	GLOBAL VENDOR NUMBER	Enter up to 7 characters.	If you selected Global above, enter the first half of the Honeywell provided IEEE vendor number.

Page 1

CABLE
FOR
OUTPUT

OUTPUT

ALT.

GLOBAL

CABLE
FOR
OUTPUT

OUTPUT

ALT.

GLOBAL

CABLE
FOR
OUTPUT

OUTPUT

ALT.

GLOBAL

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form — NG REMOTE SYSTEMS

LINE	NAME	FORM ENTRY	REMARKS
NETWORK GATEWAY—REMOTE SYSTEM ID DISPLAY			
Note —the system allows interconnection of up to 64 LCNs. The form provides space for three sets of configuration information. Make as many copies of the form as you need to configure the system.			
170	#	Enter a number from 1 to 64.	Enter the Remote System's Local System number.
	TITLE	Enter two characters. The first must be an alpha.	Enter the Remote System's Local 2-character alphanumeric title.
	DESCRIPTOR	Enter up to 20 characters.	Enter the Remote System's 20-character descriptor.
	CABLE FOR OUTPUT	Circle one: A B	Select the primary cable for sending messages to the remote system.
	RESP NG	Enter a number from 1 to 64.	Enter the LCN Node number for the NG that is responsible for routing messages to the remote system. The NG must be configured on this network.
	ALT NG	Enter a number from 1 to 64.	Enter the LCN Node number for the NG that is the alternate for routing messages to the remote system (when the responsible NG is unavailable).
	RESP LOCAL	Enter a number from 1 to 64.	Enter the PIN Local Address of the remote system's responsible NG.
	RESP GLOBAL	Enter a number— (second half of the Honeywell supplied IEEE global address for the remote node).	Enter the global vendor offset of the Remote System's responsible NG.
	ALT LOCAL	Enter a number from 1 to 64.	Enter the PIN Local Address of the Remote System's Alternate NG (optional info used to validate the source of remote messages).
	ALT GLOBAL	Enter a number— (second half of the Honeywell supplied IEEE global address for the remote node).	Enter the global vendor offset of the remote system's Alternate NG (optional info used to validate the source of remote messages).

SYSTEM-WIDE VALUES CONFIGURATION FORM

NETWORK GATEWAY (NG) SECURITY DATABASE AUTHORIZATION

190 Remote System Number | | | or ALL SYSTEMS

DEFINE ACCESS RESTRICTIONS

NETWORK GATEWAY (NG) SECURITY DATABASE DISPLAY—Page 1 of 2

SYSTEM VOLUMES/DIRECTORIES

USER VOLUMES/DIRECTORIES

191 DEFAULTS:

NO ACCESS **READ ONLY** READ/WRITE

NO ACCESS READ ONLY **READ/WRITE**

192 EXCEPTIONS:

VOL	ACCESS	VOL	ACCESS
	NO ACCESS READ ONLY READ/WRITE		NO ACCESS READ ONLY READ/WRITE
	NO ACCESS READ ONLY READ/WRITE		NO ACCESS READ ONLY READ/WRITE
	NO ACCESS READ ONLY READ/WRITE		NO ACCESS READ ONLY READ/WRITE
	NO ACCESS READ ONLY READ/WRITE		NO ACCESS READ ONLY READ/WRITE
	NO ACCESS READ ONLY READ/WRITE		NO ACCESS READ ONLY READ/WRITE
	NO ACCESS READ ONLY READ/WRITE		NO ACCESS READ ONLY READ/WRITE
	NO ACCESS READ ONLY READ/WRITE		NO ACCESS READ ONLY READ/WRITE
	NO ACCESS READ ONLY READ/WRITE		NO ACCESS READ ONLY READ/WRITE
	NO ACCESS READ ONLY READ/WRITE		NO ACCESS READ ONLY READ/WRITE

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 1 of 2)

LINE	NAME	FORM ENTRY	REMARKS
NETWORK GATEWAY—SECURITY DATABASE DISPLAY—PAGE 1			
DESIRED DEFAULT AUTHORIZATIONS			
190	REMOTE SYSTEM NUMBER or ALL	Enter a number from 1 to 64 or circle ALL	Identifies the Remote System number for which access rights are being specified. If the access restrictions apply to all systems, circle ALL. Then choose access rights and exceptions as described below.
191	DEFAULTS System/User Volumes and Directories	Circle one (each column)— No Access Read Only or Read Write	Designates default access to all System /User Volumes/Directories in the specified system(s).
192	EXCEPTIONS		
	VOL	Enter up to four characters	Specifies the Volume or directory for which the following exceptions apply.
	ACCESS	Circle one— No Access Read Only or Read Write	Specifies access allowed to the designated Volume or Directory.
	Note that the above exceptions can be specified for System Volumes/Directories (left column) and/or for User Volumes/Direc- tories (right column).		

SYSTEM-WIDE VALUES CONFIGURATION FORM

NETWORK GATEWAY (NG) SECURITY DATABASE DISPLAY-Page 2 of 2

200 Remote System Number | or ALL

PARAMETER ACCESS

201 DEFAULTS: NO ACCESS **READ ONLY** READ/WRITE

EXCEPTIONS:

203	NODE	TYPE	204	HIWAY/UNIT	205	ACCESS
AM	CM	HG	NM		NO ACCESS	READ ONLY READ/WRITE
AM	CM	HG	NM	<u> </u>	NO ACCESS	READ ONLY READ/WRITE
AM	CM	HG	NM		NO ACCESS	READ ONLY READ/WRITE
AM	CM	HG	NM	<u> </u>	NO ACCESS	READ ONLY READ/WRITE
AM	CM	HG	NM		NO ACCESS	READ ONLY READ/WRITE
AM	CM	HG	NM	<u> </u>	NO ACCESS	READ ONLY READ/WRITE
AM	CM	HG	NM	<u> </u>	NO ACCESS	READ ONLY READ/WRITE
AM	CM	HG	NM	<u> </u> <u> </u>	NO ACCESS	READ ONLY READ/WRITE
AM	CM	HG	NM	<u> </u>	NO ACCESS	READ ONLY READ/WRITE
AM	CM	HG	NM		NO ACCESS	READ ONLY READ/WRITE

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form (Page 2 of 2)

LINE	NAME	FORM ENTRY	REMARKS
NETWORK GATEWAY—SECURITY DATABASE DISPLAY—PAGE 2 OF 2			
SECURITY DATABASE EXCEPTIONS			
200	REMOTE SYSTEM NUMBER or ALL	Enter a number from 1 to 64 or circle ALL	Identifies the Remote System number for which access rights are being specified. If the access restrictions apply to all systems, circle ALL. If ALL was selected on the previous form, select ALL on this form too. Then choose access rights and exceptions as described below.
201	DEFAULTS	Circle one— No Access Read Only or Read Write	Designates default parameter access for remote Systems.
EXCEPTIONS			
203	NODE TYPE	Circle one— AM CM HG NM for each Hiway/Unit where you want to restrict access.	Identifies the Node Type for which an exception is authorized.
204	HIWAY/UNIT	Enter up to two characters for the Hiway Number or Unit <u>ID</u> .	Designates the Hiway/Unit on which the Node type (above) is configured.
205	ACCESS	Circle one of the following: NO ACCESS READ ONLY READ/WRITE	Designates access to the specified Node and Hiway/Unit (above).

SYSTEM-WIDE VALUES CONFIGURATION FORM

NETWORK GATEWAY (NG) MODEM DEFINITION

Page 1

WARNING: DO NOT CHANGE VALUES ON THIS PAGE UNLESS REQUESTED TO DO SO BY HONEYWELL

230	HIGH PRIORITY TARGET HOLD TIME	128
231	PRIORITY 1 TARGET ROTATION TIME	8190
232	PRIORITY 2 TARGET ROTATION TIME	6800
233	PRIORITY 3 TARGET ROTATION TIME	5000
234	TARGET ROTATION TIME RING MAINT.	3500
235	RING MAINT. INITIAL	7500
236	MAX. INTER-SOLICIT	250
237	SLOT TIME	625
238	MAX. RETRY	3
239	NOISE THRESHOLD	3
240	FCS THRESHOLD	3
241	RM THRESHOLD	3
242	FRAGMENT THRESHOLD	3
243	PREAMBLE AFTER SILENCE	5
244	PREAMBLE BETWEEN	20
245	MODEM FREQUENCY	2
246	ACCEPT ALL NODES	1
247	FULL TOKEN STATISTICS	0
248	TOKEN ROTATION	9999
249	TOKEN PASS/FAIL	5
250	STATISTICS RESET	0
251	MODEM CONTROL REGISTER	0
252	MULTICAST MASK (LOW)	768
253	MULTICAST MASK (MED)	0
254	MULTICAST MASK (HIGH)	0

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form —NG MODEM DEFINITION

LINE	NAME	REMARKS
NETWORK GATEWAY MODEM DISPLAY —RESERVED VALUES		
Warning —the following data is provided for reference. Do not change values unless instructed by Honeywell. These values are typical and may change with subsequent software/hardware changes.		
230	High Priority Target Hold Time	Maximum time frames from highest priority queue may be transmitted.
231	Priority 1 Target Rotation Time	Length of time that frames from priority queue 1 may be transmitted.
232	Priority 2 Target Rotation Time	Length of time that frames from priority queue 2 may be transmitted.
233	Priority 3 Target Rotation Time	Length of that time frames from priority queue 3 may be transmitted.
234	Target Rotation Time Ring Maint.	Maximum timer value for ring maintenance (octets).
235	Ring Maint. Initial Value	Initial Value for Ring maintenance timer.
236	Max. Inter-Solicit Count	Number of token passes before a node invites other nodes into the token ring (octets).
237	Slot Time	Maximum delay to get an immediate response from a node (includes cable delay).
238	Max. Re-try	Maximum number of times that the TBC tries to transmit a frame (range 0-7).
239	Noise Threshold	Maximum delay to get an immediate response from a node (includes cable delay).
240	FCS Threshold	Maximum number of allowable frame checksum sequence error counts within 300 msec.
241	RM Threshold	Maximum number of allowable remodulation error counts within 300 msec.
242	Fragment Threshold	Maximum number of allowable frame fragment error counts within 300 msec.
243	Preamble After Silence	Number of preamble octets to be transmitted before a frame after silence.
244	Preamble Between Frames	Number of preamble octets to be transmitted between frames (range 3-32).
245	Modem Frequency	Receive/Transmit frequency pairs for the Modem range 0-4.

(Continued)

**Table 3-13 — Instructions for System-Wide Values Configuration Form —NG MODEM DEFINITION
(Continued)**

LINE	NAME	REMARKS
NETWORK GATEWAY MODEM DISPLAY —RESERVED VALUES		
Note —the following data is provided for reference. Do not change values unless instructed by Honeywell. These values are typical and may change with subsequent software releases.		
246	Accept All Nodes	0 = Only accept message from configured PIN address, or 1 = accept messages from any PIN address.
247	Full Token Statistics	0 = limited network statistics, If no token, pass counts or 1 = full statistics.
248	Token Rotation Threshold	Maximum time for a full token rotation on the network.
249	Token Pass/Fail Threshold	Maximum number of token pass failures within 300 msec (range = 0-100).
250	Statistics Reset Period	Maximum time for a full token rotation of the network.
251	Modem Control Register	Vendor-specific Modem Control Register value.
252	Multicast Mask (Low)	Value of low order bits (0-15) of the 48-bit Multicast Mask.
253	Multicast Mask (Med)	Value of medium order bits (16-31) of the 48-bit Multicast Mask.
254	Multicast Mask (High)	Value of high order bits (32-48) of the 48-bit Multicast Mask.

SW88-615

SYSTEM-WIDE VALUES CONFIGURATION FORM

NETWORK GATEWAY (NG) ERROR MESSAGE THROTTLE (R520 and later)

300 NUMBER OF ERROR MESSAGES ALLOWED PER 4 HOUR INTERVAL
default = 0

TAG NAME OPTIONS Page 1

320 TAG NAME SIZE SHORT LONG

322 FORMAT ON YES NO

AM MULTIPLE PRIMMOD ALARMING OPTIONS (R520 and later) Page 1

340 AM MULTIPLE DISABLED ENABLED EXCLUSIVE ENABLED
MODE INCLUSIVE
MODE MODE

HG OPTIONS Page 1

360 HG SP/OP CLAMP OPTION DISABLED OPTION ENABLED
OPTION OPTION

361 SYSTEM ALARM ON DHP/PIU POINT LOAD INHIBIT ENABLE
(R680 and later)

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-13 — Instructions for System-Wide Values Configuration Form

LINE	NAME		REMARKS
NG ERROR MESSAGE THROTTLE			
300	NUMBER OF ERROR MESSAGES ALLOWED PER 4 HOUR INTERVAL	Enter a number from 0 to 32767 (R520 and later)	Noise on external networks may generate false error messages.
TAG NAMES OPTIONS			
320	TAG NAME SIZE	Circle one: SHORT LONG	Select SHORT for 8-character tag names; LONG for 16 character tag names.
322	FORMATTING ON LONG TAG NAME	Circle one: YES NO	Select YES to apply formatting to tag names. Refer to the Section 3 in the <i>Data Entity Builder</i> manual for information.
AM MULTIPLE PRIMMOD ALARMING OPTIONS			
340	AM MULTIPLE PRIMMOD ALARMING	Circle one: Disabled or Enabled Inclusive Mode or Enabled Exclusive Mode (R520 and later)	Disabled = alarms reflected only in Primmod parameter alarm group. Enabled Exclusive Mode = Primmod and \$MPROD alarms are mutually exclusive. Enabled Inclusive = alarms go to both Primmod and \$MPROD alarm groups.
HG OPTIONS			
360	HG SP/OP CLAMP OPTION	CIRCLE ONE: DISABLE OPTION or ENABLE OPTION (R600 and Later)	DISABLE OPTION = will not honor SP and OP limits (existing functionality) ENABLE OPTION = will HONOR SP and OP Limits
361	SYSTEM ALARM ON DHP/PIU POINT LOAD	SELECT ONE: INHIBIT OPTION (R680 and later) or ENABLE OPTION	INHIBIT OPTION = suppresses the system alarm, when the Hiway box state changes are caused by the point load or change of address related parameters of DHP or PIU. ENABLE OPTION = enables the system alarm, when the Hiway box state changes are caused by the point load or change of address related parameters of DHP or PIU. Refer to the Section 13.7 in the <i>Process Operations Manual</i> for information.

HM VOLUME CONFIGURATION FORM

VOLUME CONFIGURATION MENU

1 NODE PAIR NUMBER | | | SELECT VOLUME(S) FOR THIS HM

2 NUMBER OF WINCHESTER DISKS | |
default = 1

PROGRAM IMAGE

3 DISK TYPE/SIZE | | | | | | | |
default = 1

AREA

4 HM INIT PERSONALITY YES NO

CHECKPOINT

5 SYSTEM VOLUME YES NO

CONTINUOUS HISTORY

6 GDF FILES INCLUDED YES NO

JOURNALS

7 FILE MANAGER DESCRIPTORS YES NO

CL STORAGE

8 AREA SOE JOURNAL SIZE | | | |
default = 0

USER FILE STORAGE

9 BURST BUFFER SIZE
FOR PROCESS/SYSTEM JOURNALS | | | |
default = 1500

10 SYSTEM UNIT JOURNALS? YES NO

10a STATUS CHANGE ERROR MAINT

| | | | | | | | | | | | | | | |
default = 0 0 0

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

3.14 HM VOLUME CONFIGURATION FORM SW88-617

Table 3-14 contains instructions for Configuration Form SW88-617.

Table 3-14 — Instructions for HM Volume Configuration Form (Page 1 of 9)

LINE	NAME	FORM ENTRY	REMARKS
VOLUME CONFIGURATION MENU DISPLAY			
1	NODE PAIR NUMBER	Enter a number from 1 to 20.	
2	NUMBER OF WINCHESTER DISKS	Enter 1 or 2.	Enter the number of Winchester disk drives in this HM.
3	DISK TYPE/SIZE	Enter 3, 4, 5, 6, or 7. Enter formatted size in KB if SCSI compatible disk.	3 = Wren 3 (136900 Kb/disk). SCSI, type; formatted sizes: 4 = 215040 Kb (210 MB) 5 = 454130 Kb (445 MB) 6 = 896046 Kb (875 MB) 7 = 1836.736 Mb (1.8 G)
4	HM INIT PERSONALITY	Circle one: YES NO	Select YES to configure the HM Off-Line Personality on the local volume.
5	SYSTEM VOLUME	Circle one: YES NO	Assigns the System Files to this HM.
6	GDF FILES INCLUDED	Circle one: YES NO	Circle YES to allocate storage space for GDF files on this HM.
7	FILE MANAGER DESCRIPTORS	Circle one: YES NO	Select YES if you want files in the system volume to have File Descriptors (requires 4 Kb + 64 bytes/file).
8	AREA SOE JOURNAL SIZE	Enter up to 5 numeric digits. Default =0 (no journal).	No. of events can range from 200 to 15,000. Size should ≥ twice expected no. of events. <u>Important - Also see Journals, Process Unit Journal Configuration</u> (lines 80/81).
9	BURST BUFFER SIZE	Enter up to 4 numeric digits. Default =400.	The size of the burst buffer (number of events) can range from 400 to 7500.
10	SYSTEM UNIT JOURNALS?	Circle one: YES NO	Circle YES if this HM is to include System Journaling. Then fill in 10a, 10b, and 10c.
10a	STATUS CHANGE	Enter up to 5 numeric digits.	The number of events can range from 200 to 12,500.
	ERROR	Enter up to 5 numeric digits.	The number of events can range from 200 to 15,000.
	MAINT	Enter up to 5 numeric digits.	The number of events can range from 200 to 15,000.
NOTE Refer to the <i>Engineer's Reference Manual</i> for more information on volume size.			

HM VOLUME CONFIGURATION FORM (Continued)

PROGRAM IMAGE VOLUME CONFIGURATION - SELECT NODE DISPLAY PAGE 1

30	NODE	PAIR	NUMBER			
1	1	2	1			
2	1	2	2			
3	1	2	3			
4	1	2	4			
5	1	2	5			
6	1	2	6			
7	1	2	7			
8	1	2	8			
9	1	2	9			
10	1	2	10			
11	1	2	11			
12	1	2	12			
13	1	2	13			
14	1	2	14			
15	1	2	15			
16	1	2	16			
17	1	2	17			
18	1	2	18			
19	1	2	19			
20	1	2	20			
21	1	2	21			
22	1	2	22			
23	1	2	23			
24	1	2	24			
25	1	2	25			
26	1	2	26			
27	1	2	27			
28	1	2	28			
29	1	2	29			
30	1	2	30			
31	1	2	31			
32	1	2	32			
33	1	2	33			
34	1	2	34			
35	1	2	35			
36	1	2	36			
37	1	2	37			
38	1	2	38			
39	1	2	39			
40	1	2	40			
41	1	2	41			
42	1	2	42			
43	1	2	43			
44	1	2	44			
45	1	2	45			
46	1	2	46			
47	1	2	47			
48	1	2	48			
49	1	2	49			
50	1	2	50			
51	1	2	51			
52	1	2	52			
53	1	2	53			
54	1	2	54			
55	1	2	55			
56	1	2	56			
57	1	2	57			
58	1	2	58			
59	1	2	59			
60	1	2	60			
61	1	2	61			
62	1	2	62			
63	1	2	63			
64	1	2	64			
65	1	2	65			
66	1	2	66			
67	1	2	67			
68	1	2	68			
69	1	2	69			
70	1	2	70			
71	1	2	71			
72	1	2	72			
73	1	2	73			
74	1	2	74			
75	1	2	75			
76	1	2	76			
77	1	2	77			
78	1	2	78			

```
31 SELECT ALL PERSONALITIES TO RESIDE ON THIS HM:
   (QLT AND OFT PERSONALITIES ARE INCLUDED IF ANY PERSONALITY
    IS CONFIGURED)
```

CM II HG II AM II USOPR II USUNP II NIM II NG II

CM IV HG IV AM IV USOPR IV USUNP IV NIM IV NG IV

[illegible]

AREA DATA VOLUME CONFIGURATION DISPLAY PAGE 1

40	NODE PAIR NUMBER				41	FILE MANAGER DESCRIPTORS	YES	NO
----	------------------	--	--	--	----	--------------------------	-----	----

42

AREA	NAME	VOLUME	SIZE	NUMBER
		(KB)		FILES

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-14 — Instructions for HM Volume Configuration Form (Page 2 of 9)

LINE	NAME	FORM ENTRY	REMARKS
PROGRAM IMAGE VOLUME CONFIGURATION SELECT NODE TYPES DISPLAY			
30	NODE PAIR NUMBER	Enter a number from 1 to 20.	Same as page 1, line 1.
31	SELECT ALL NODE TYPES THAT APPLY	Circle items that apply CM II, CM IV HG II, HG IV AM II, AM IV USOPR II, USOPR IV USUNP II USUNP IV NIM II, NIM IV NG II, NG IV	Allocates space on this HM for the selected program images. Modules that contain an HMPU, HPK2, or K2LCN board are version II (Check card slots 1 and 7). For modules that contain a K4LCN board, select version IV.
33	HIST MOD DUMP VOL	Circle if HM Dump Volume is to be configured on this History Module.	
	DUMP VOLUME SIZE (KB)	Enter 0 for no dump volume. Enter a number from 7000 to maximum disk size if you want a dump volume for any node type.	Dump-volume size (in Kb) can range from 0 to maximum disk size. Suggested number is equal to the memory size for the module with the largest memory.
AREA DATA VOLUME CONFIGURATION DISPLAY			
40	NODE PAIR NUMBER	Enter a number from 1 to 20.	Same as page 1, line 1.
41	FILE MANAGER DESCRIPTORS	Circle one: YES NO	Circle YES if you want files in this volume to have File Descriptors (requires 4 Kb/ volume plus 64 bytes/file).
42	AREA NAME	For each of up to 10 areas, enter up to 8 characters.	Area names must be entered exactly as they were entered in the Area Names Area Descriptors Configuration Form. (See Table 3-2.)
	VOLUME SIZE (KB)	For each of up to 10 areas, enter 1 to 5 numeric digits.	Volume size ranges from 1 to max disk size in Kb.
	NUMBER FILES	For each of up to 10 areas, enter up to 3 numeric digits.	The number of files ranges from 0 to 999.

HM VOLUME CONFIGURATION FORM (Continued)

PHYSICAL NODE FOR CHECKPOINT CONFIGURATION DISPLAY

PAGE N

50 NODE PAIR NUMBER

51	NODE	VOLUME	NUMBER	NODE	VOLUME	NUMBER
	NO.	SIZE	FILES	NO.	SIZE	FILES

[illegible]

A 10x20 grid of small vertical bars, resembling a barcode or a data visualization. The bars are arranged in 10 rows and 20 columns, with varying heights and positions, creating a complex pattern.

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-14 — Instructions for HM Volume Configuration Form (Page 3 of 9)

LINE	NAME	FORM ENTRY	REMARKS
PHYSICAL NODE FOR CHECKPOINT CONFIGURATION DISPLAY			
50	NODE PAIR NUMBER	Enter a number from 1 to 20.	Refer to Table 3-12.
51	NODE NO	Enter either 0 or a number from 1 to 64.	Enter physical number for a valid node type. Valid node types are AM, CM, HG, or NIM. Enter 0 if node is not configured or 1 to 64 to configure a node for checkpointing. Redundant node numbers must be entered one after the other.
	VOLUME SIZE	Enter appropriate number per remarks column.	Enter 0 for HGs and redundant members. Refer to the <i>Engineer's Reference Manual</i> for guidance on other volume sizes.
	NUMBER FILES	Enter either 0 or a number within the appropriate range.	Must be 0 for HG and redundant members. For each AM , enter $5 \times (\text{number of units} + 2)$. The range is 10 to 506. For each CM , enter $4 \times (\text{number of Units} + 1)$. The range is 8-404. For each NIM , enter $4 \times (\text{No. of PMS/APMs/HPMs}) + 4 \times (\text{No. of LMs}) + 4$. The range is 1 to 999.

Table 3-14 — Instructions for HM Volume Configuration Form (Page 4 of 9)

LINE	NAME	FORM ENTRY	REMARKS
CONTINUOUS HISTORY UNIT AND GROUPS DISPLAY			
60	NODE PAIR NUMBER	Enter one or two numeric digits.	Refer to Table 3-12.
61	UNIT ID	Enter two alphanumeric characters for each of up to 64 units.	The Unit ID must be entered exactly as it is entered on the Unit ID/Unit Descriptor Form. (See Table 3-1.)
	NBR. GROUPS	The number of groups ranges from 1 through 150.	The number of groups ranges from 1 through 150. A maximum of 120 groups is recommended to avoid over-loading the HM. The number of groups can safely exceed 120 if the number of snap-shots selected is few or none. Each group contains up to 20 points. The sum of numbers in all entries must not exceed 120.

**HM VOLUME CONFIGURATION FORM
(Continued)**

CONTINUOUS HISTORY HM/UNIT OPTIONS	PAGE N
------------------------------------	--------

65 NODE PAIR NUMBER

66 ALL UNITS

	SAVE RATE	SAVE OPTIONS	PREARCHIVE HOURS SNAPSHOTS	USER AVG	ARCHIVE?
05 10 20 60		SNAPSHOTS USER AVG			YES NO

OR

67 DESIGNATED UNIT	SAVE RATE	SAVE OPTIONS	PREARCHIVE HOURS SNAPSHOTS	USER AVG	ARCHIVE?
--------------------------	--------------	--------------	-------------------------------	----------	----------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
---	-------------	--------------------	---	---	---------------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
---	-------------	--------------------	---	---	---------------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
---	-------------	--------------------	---	---	---------------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
---	-------------	--------------------	---	---	---------------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
---	-------------	--------------------	---	---	---------------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
---	-------------	--------------------	---	---	---------------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
---	-------------	--------------------	---	---	---------------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
---	-------------	--------------------	---	---	---------------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
---	-------------	--------------------	---	---	---------------

	05 10 20 60	SNAPSHOTS USER AVG			YES NO
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References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

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Table 3-14 — Instructions for HM Volume Configuration Form (Page 5 of 9)

CONTINUOUS HISTORY HM/UNIT OPTIONS DISPLAY			
NOTE			
This form can be used to mass-configure all options for <u>all</u> Units or all options for <u>specific</u> Units. Option selections on this page override any entry on Page 6 of this form for the same Unit. If you choose ALL UNITS, do not use Page 6 of this form. If you configure the options for a DESIGNATED UNIT here, do not name that Unit again on Page 6. You can leave Page 5 blank or choose options for some Units here and choose the more detailed options for other Units on Page 6.			
LINE	NAME	FORM ENTRY	REMARKS
65	NODE PAIR NUMBER	Enter one or two digits	
66	ALL UNITS	Circle if desired	If you select ALL UNITS, choose the options for all Groups and Units on the HM.
	SAVE RATE	Circle rate (seconds)	See Description Below
	SAVE OPTIONS	Circle if desired	See Description Below
	PREARCHIVE HOURS	Enter one to three digits	See Description Below
	ARCHIVE?	Circle one: YES or NO	See Description Below
67	DESIGNATED UNIT	Enter the Unit number (Not applicable if you circled ALL UNITS)	If you designate a Unit, the options entered apply to all Groups on that Unit.
	SAVE RATE	Circle the collection frequency: 5, 10, 20, or 60 seconds. (you must select a Save Rate to get <u>any</u> history)	Recommended snapshots/hour maximum = 168. Maximum allowed at 5 sec. = 240, 10 sec. = 480, 20 sec. = 960, 60 sec. = 2880.
	SAVE OPTIONS	For each Designated Unit configured, circle:	Select the type of data to be saved for all Groups in the Designated Unit.
		SNAPSHOTS Trends require snapshots to be selected.	–base history plus 8 hours of 1-minute averages
		USER AVG (User Average)	–base history plus 80 user averages
		SNAPSHOTS & USER AVG	–base history plus snapshots plus user averages
		Neither Snapshots nor User Averages	–only "base" history (hourly, shift, daily, monthly) averages
	PREARCHIVE HOURS USER AVG SNAPSHOTS	As applicable, enter one to four numeric digits for User Average and/or Snapshots.	Prearchive hours (either User Average or Snapshots can range from 0 to 7,496). If you want more than 8 hours of data you must use prearchiving. (This is the way to exceed the standard 8 hours of 1-minute snapshots and 85 user averages.)
	ARCHIVE	Circle one: YES NO (not currently available)	Select YES if all Groups in this Unit are to be archived.

HM VOLUME CONFIGURATION FORM (Continued)

CONTINUOUS HISTORY GROUP OPTIONS	PAGE N
----------------------------------	--------

70 NODE PAIR NUMBER

71

UNIT ID	GROUP NO.	SAVE RATE	SAVE OPTIONS	PREARCHIVE HOURS	ARCHIVE?
				SNAPSHOTS USER AVG	
		05 10 20	60 SNAP-SHOTS USER AVG		YES NO
		05 10 20	60 SNAP-SHOTS USER AVG		YES NO
		05 10 20	60 SNAP-SHOTS USER AVG		YES NO
		05 10 20	60 SNAP-SHOTS USER AVG		YES NO
		05 10 20	60 SNAP-SHOTS USER AVG		YES NO
		05 10 20	60 SNAP-SHOTS USER AVG		YES NO
		05 10 20	60 SNAP-SHOTS USER AVG		YES NO
		05 10 20	60 SNAP-SHOTS USER AVG		YES NO
		05 10 20	60 SNAP-SHOTS USER AVG		YES NO

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
 Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

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Table 3-14 — Instructions for HM Volume Configuration Form (Page 6 of 9)

LINE	NAME	FORM ENTRY	REMARKS
CONTINUOUS HISTORY GROUP OPTIONS DISPLAY			
70	NODE PAIR NUMBER	Enter one or two numeric digits.	Refer to Table 3-12.
71	UNIT ID	Enter two alphanumeric characters.	Each Unit ID entered on this form can be one of up to 60 entered on Form SW88-506.
	GROUP NO.	Enter a number from 1 to 120.	The Group Number can vary from 1 to 120. Groups are numbered from 1 to N for each Unit, where N is the number of groups.
	SAVE RATE	Circle the collection frequency: 5, 10, 20, or 60 seconds.	Recommended snapshots/ hour maximum = 168. Maximum allowed at 5 sec. = 240, 10 sec. = 480, 20 sec. = 960, 60 sec. = 2880.
	SAVE OPTIONS	For each Group Number configured above, circle:	Select the type of data to be to be saved for the Group specified:
		SNAPSHOTS Trends require snapshots to be selected.	—base history plus 8 hours of 1-minute averages.
		USER AVG (User Average)	—base history plus 80 user averages.
		SNAPSHOTS & USER AVG.	—base history plus snapshots plus user averages.
	PREARCHIVE HOURS USER AVG SNAPSHOTS	Neither Snapshots nor User Averages.	—only "base" history (hourly, shift, daily, monthly) averages.
		As applicable, enter one to four numeric digits for User Average and/or Snapshots.	Prearchive hours (either User Average or Snapshots can range from 0 to 7,496). If you want more than 8 hours of data you must use prearchiving. (This is the way to exceed the standard 8 hours of 1-minute snapshots and 85 user averages.)
	ARCHIVE	Circle one: YES* NO	Select YES if this Group is to be archived.
*Not currently available			

HM VOLUME CONFIGURATION FORM
(Continued)

PROCESS UNIT JOURNAL CONFIGURATION

PAGE N

80 NODE PAIR NUMBER |__|

81

[illegible]

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-14 — Instructions for HM Volume Configuration Form (Page 7 of 9)

LINE	NAME	FORM ENTRY	REMARKS
PROCESS UNIT JOURNAL CONFIGURATION DISPLAY			
80	NODE PAIR NUMBER	Enter one or two numeric digits.	Node pair ranges from 1 to 20.
81	UNIT ID	For each of up to 36 Units that are to be Journalled: – enter two alphanumeric characters	The Unit ID must be entered exactly as it is entered on the Unit ID/Unit Descriptor Form. (See Table 3-1.) The Unit ID SY (reserved in the system) must not be used. Note that the area SOE Journal collects events from only the units specified here. If an Area SOE Journal, was configured (line 8), a Process Unit Journal must be configured on the <u>same</u> HM node pair.
	PROCESS ALARM	– enter up to five numeric digits	The number of events entered must range from 200 to 15,000.
	PROCESS CHANGE	– enter up to five numeric digits	The number of events entered must range from 200 to 10,000.
	OPERATOR MESSAGE	– enter up to four numeric digits	The number of events entered must range from 200 to 5000.

HM VOLUME CONFIGURATION FORM (Continued)

CL STORAGE VOLUME PAGE 1

110 NODE PAIR NUMBER | | | 111 HIWAY FILE MANAGER YES NO

112 NETWORK FILE MANAGER DESCRIPTORS YES NO

113 HIWAY NETWORK	NUMBER OF FILES	VOLUME SIZE (KB)	HIWAY NETWORK	NUMBER OF FILES	VOLUME SIZE (KB)
_	_	_ _	_	_	_ _
_	_	_ _	_	_	_ _
_	_	_ _	_	_	_ _

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-14 — Instructions for HM Volume Configuration Form (Page 8 of 9)

LINE	NAME	FORM ENTRY	REMARKS
CL STORAGE VOLUME CONFIGURATION DISPLAY			
110	NODE PAIR NUMBER	Enter one or two numeric digits.	Assigns CL volumes for HG and/or NIM nodes.
111	HIWAY FILE MANAGER DESCRIPTORS	Circle one: YES NO	Select YES if you want files in this volume to have File Descriptors (requires 4Kb per volume + 64 bytes/file).
112	NETWORK FILE MANAGER DESCRIPTORS	Circle one: YES NO	Select YES if you want files in this volume to have File Descriptors (requires 4Kb per volume + 64 bytes/file).
113	HIWAY NETWORK	Enter one to two numeric digits.	The hiway or network number ranges from 0 to 20. If not 0, it must be assigned.
	NUMBER OF FILES	Enter one to three numeric digits.	The number of files should be one for each CL sequence program. The range is 1 to 999.
	VOLUME SIZE (KB) SIZE	Enter one to four numeric digits.	Volume Size can range from 1 to maximum disk size in Kb.

HM VOLUME CONFIGURATION FORM

(Continued)

```
USER  FILE  STORAGE  CONFIGURATION  DISPLAY
```

PAGE 1

130	NODE PAIR NUMBER			
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131

VOLUME ID	NUMBER OF FILES	VOLUME SIZE KILOBYTE	FILE MANAGER DESCRIPTORS
<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u> <u> </u> <u> </u>	YES FAST
			YES NO
<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u> <u> </u> <u> </u>	YES NO
			YES NO
			YES NO
			YES NO
			YES NO
<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u> <u> </u> <u> </u>	YES NO
			YES NO
<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u> <u> </u> <u> </u>	YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u> <u> </u> <u> </u>	YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
			YES NO
<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u>	<u> </u> <u> </u> <u> </u> <u> </u> <u> </u> <u> </u>	YES NO
			YES NO

References: Network Form Instructions in the Implementation/Startup & Reconfiguration - 1 binder.
Network Data Entry in the Implementation/Startup & Reconfiguration - 1 binder.

Table 3-14 — Instructions for HM Volume Configuration Form (Page 9 of 9)

USER FILE STORAGE CONFIGURATION DISPLAY			
Refer to the <i>Engineer's Reference Manual</i> for information on selecting volume size.			
LINE	NAME	FORM ENTRY	REMARKS
130	NODE PAIR NUMBER	Enter 1 or 2 numeric digits.	Refer to Table 3-12.
131	VOLUME ID	Enter up to 4 characters.	The Virtual Volume ID must include one alpha character. Underbar can appear in position 2 or 3, but not both.
	NUMBER OF FILES	Enter up to 4 numeric digits.	The number of files can range from 0 to 9995.
	VOLUME SIZE (KILOBYTES)	Enter up to 5 digits.	Volume size can range from 0 to maximum disk size in kilobytes. The sum of all volume sizes must not exceed that limit.
	FILE MANAGER DESCRIPTORS <u>First Volume</u>	Circle one: YES FAST	Select FAST if you want this to be a fast access volume. Only the first volume has this feature, so you may want to make it larger than the others to take advantage of this feature. Select YES if you want files in this volume to have File Descriptors (with normal access speed).
	FILE MANAGER DESCRIPTORS Except First Volume	Circle one: YES NO	Select YES if you want files in this volume to have File Descriptors (requires 4KB per volume + 64 bytes/file).

4 INDEX

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